The Economic Effects of Tensions in Energy Transportation*

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Abstract

This paper tackles the urgent issue of how tensions in energy transportation impact oil markets and the global economy. The study introduces an *energy transportation uncertainty* index, developed from over 50 global newspapers, to monitor fluctuations in uncertainty associated with significant events such as the Tanker War, major US sanctions, the Exxon Valdez oil spill, the Nord Stream sabotage, the Colonial pipeline cyberattack, and various Gulf incidents. Using a structural vector autoregression (SVAR) model, the analysis shows that these fluctuations in transportation uncertainty cause increases in real oil prices due to supply chain challenges. While oil production dips initially due to perceived risks, it quickly rebounds, though inventories are heavily used. These shocks also heighten geopolitical tensions and reduce global industrial output.

JEL Codes: C32, D80, E32, F51, Q43, N70, L95

Keywords: Supply chain disruptions, Energy transportation uncertainty, Geopolitical risks, Oil markets, Uncertainty shocks, SVAR

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1 Introduction

In recent years, the stability of global energy supply chains has come under unprecedented threats by high-profile incidents such as the Nord Stream sabotage and the Colonial Pipeline cyberattack, as well as the rising threats of drone warfare and piracy. These disruptions not only highlighted a critical vulnerability within our interconnected global economy but also garnered significant attention from news media and policymakers from a geopolitical perspective. Despite this widespread attention, empirical research concerning the macroeconomic and oil market impacts of such uncertainties remains limited. This paper aims to address this gap by investigating how disruptions in energy transportation impact global oil markets and macroeconomic stability. Through this research, the study seeks to improve our understanding of these dynamics and provide insights that can help policymakers navigate a landscape influenced by technological advancements and geopolitical tensions.

Related literature and contribution. This paper contributes to three distinct strands of the literature. Initially, it builds upon the foundational research by Baker et al. (2016), which investigates the macroeconomic effects of economic policy uncertainty shocks, highlighting that uncertainty generally adversely impacts economic activity and financial stability. This theme is further explored by subsequent studies, such as Husted et al. (2020) which analyzes the impact of uncertainty surrounding Federal Reserve policy actions, and Ahir et al. (2022) which introduces the World Uncertainty Index (WUI). Moreover, Caldara et al. (2020) and Caldara and Iacoviello (2022), expand this discussion to include the economic effects of trade policy uncertainty (TPU) and the measurement of geopolitical risk, respectively, which align closely with this paper's emphasis on energy transportation uncertainty.

The major contribution of this paper is the introduction of a text-based measure of energy transportation uncertainty, mined from over 50 English-language media sources

across the globe. This index captures significant events such as the Tanker War and recent incidents like the Nord Stream sabotage, providing a robust tool for analyzing the impacts of these uncertainties. The news-based ETU index allows us to analyze news articles to understand the contextual factors influencing major shock episodes and their impact in particular historical moments using variance and historical decomposition exercises.

From a methodological standpoint, my approach builds on existing textual analyses that explore oil markets to refine the quantification of uncertainty in energy markets, particularly focusing on the midstream sector. For instance, Abiad and Qureshi (2023) developed a news-based index of oil price uncertainty (OPU) using textual analysis techniques and provided a detailed examination of the macroeconomic effects of this uncertainty. Plante (2019) used a news-based OPEC index to study how oil price volatility. Similarly, Dang et al. (2023) developed an energy-related uncertainty index (the EUI) by applying textual analysis to the monthly country reports from the Economist Intelligence Unit.

The third branch relates to the fast-growing literature on how supply chain disruptions affect the economy, as explored by Benigno et al. (2022), Khalil and Weber (2023), Finck et al. (2024), Elsayed et al. (2023), Laumer (2023), Ascari et al. (2024), Carrière-Swallow et al. (2023) and Morão (2025). However, this literature primarily focuses on shipping costs and the recent surge in inflation. De Santis (2024) bridges the gap between this literature and energy supply issues by showing that in the Euro Area, core prices and inflation expectations are more influenced by supply chain disruptions than by adverse energy supply shocks. Nevertheless, a significant gap persists in empirical research concerning threats to energy supply chains, such as pipelines, terminals, and ships.

Preview of results. Energy transportation uncertainty shocks have economically significant effect. Negative news about disruptions in the energy supply chain results in a significant increase in real oil prices as demand surpasses supply due to logistical issues within the supply chain. World oil production initially drops due to perceived risks,

but it quickly rebounds. However, despite this quick restoration of production, oil inventories are drawn down to stabilize consumption levels. The shock also impacts the broader economy by escalating geopolitical tensions and leading to a significant and persistent decrease in world industrial production. The paper also investigates the historical role of energy transportation uncertainty in influencing fluctuations in key variables, using variance and historical decomposition analysis. A major finding is that news about energy transportation uncertainty substantially influenced historical fluctuations in oil prices and geopolitical risk during major episodes of turmoil in global markets such as in the 1980's the U.S. sanctions on the Soviet pipeline in Europe and the Tanker War, and more recent events such as the 2022 Russian invasion of Ukraine and the sabotage of the Nord Stream pipeline.

Furthermore, this paper evaluates the impact of omitting the year 2022 and the post-COVID-19 period from the sample to assess the effects of the pandemic and the Russian invasion of Ukraine in the main results. Another test is positioning the index at the forefront of the recursive ordering in the SVAR model. Additionally, the study conducts a comprehensive series of sensitivity checks to ensure the robustness of these findings. These checks include various factors such as the inclusion or exclusion of exogenous variables, the priors used in the estimation process, and different model specifications, including the number of lags.

Roadmap. In what follows, Section 2 provides background on the energy transportation industry. Section 3 describes the index and analyzes the major events captured by it. Section 4 outlines the empirical model, data, and identification method used to estimate the impacts of transportation uncertainty. Section 5 examines the effects and historical significance of energy transportation uncertainty shocks. Section 6 presents additional results using different specifications and several series of sensitivity checks. Finally, Section 7 offers policy implications and concluding remarks.

2 Institutional background

The transportation of energy resources began with the Industrial Revolution, initially relying on coal transported by wagons and flatboats. However, by the 1860s, the oil industry required more efficient methods. This need led to the development of the first pipelines, which enabled the transportation of large quantities of oil and allowed private companies to control its transport, marking a significant technological advancement. The post-World War II expansion of global trade and industrialization led to an increased demand for oil and gas, prompting the construction of extensive pipeline networks across industrialized countries, and the development of large-scale tanker ships.

Transportation assets. There are four main ways to transport energy resources. Pipelines are a key infrastructure and have become the preferred choice for efficiently and cost-effectively moving large volumes of crude oil and natural gas over long distances. On the other hand, rail and truck transportation offer greater flexibility in terms of timing and routes, especially where pipelines are neither practical nor economical. For instance, most of the Bakken shale oil in North Dakota relies on rail transport due to limited pipeline infrastructure. Trucks are typically employed for shorter distances. For international transport, oil tankers, including Very Large Crude Carriers (VLCCs) and Ultra Large Crude Carriers (ULCCs), carry oil across oceans. Kilian et al. (2023) provide an insightful analysis of the VLCC market between the Arabian Gulf and the Far East. Another key infrastructure is storage facilities, which include terminals and storage tanks critical for managing supply and demand fluctuations. In crude oil logistics, oil is initially moved through small pipelines to a central location with sufficient storage volume to be efficiently shipped via truck, major pipelines, barge, or rail.

Natural gas, unlike crude oil, cannot be stored near the production site. Instead, it is transported to a processing facility to remove impurities and water before being sent to underground storage, such as depleted oil and natural gas fields, salt formations and de-

pleted aquifers. Recent advancements in Liquefied Natural Gas (LNG) technology have significantly facilitated the large-scale international trade of natural gas in liquid form, utilizing facilities such as liquefaction plants, regasification terminals, and specialized LNG carriers.

Although this paper does not focus on them, there is also power grid infrastructure that includes generation plants, substations, and distribution networks, where electricity is transported via high-voltage power lines. Recently, High Voltage Direct Current (HVDC) lines have been used for long-distance and underwater electricity transmission, offering higher efficiency than traditional AC systems.

Key players. In the pipeline sector, midstream companies in the US have become more specialized, with lesser-known names like Enbridge, TransCanada, Kinder Morgan emerging as major players. Globally, the Russian company Transneft is a significant entity in this sector.

Similar to the upstream sector, national oil companies (NOCs) also play a significant role by controlling large portions of their countries' oil infrastructure. Examples include Saudi Aramco in Saudi Arabia, Petrobras in Brazil, and CNPC in China. Major international oil companies (IOCs), such as ExxonMobil, Shell, BP, and Chevron, are typically vertically integrated, operating extensive pipeline networks and shipping fleets. In the international natural gas markets, Gazprom in Russia and Qatar Petroleum are key NOCs, while Shell and ExxonMobil are major IOCs.

Risk management. Banks and financial institutions are key players in this market, offering customized financial products. Commodity exchanges like the New York Mercantile Exchange (NYMEX) and Intercontinental Exchange (ICE) provide platforms for trading derivatives essential for hedging price risks associated with energy transportation. These can significantly impact costs, revenues, and market stability.

Firms involved in energy transportation use hedging strategies and insurance to mit-

igate these risks. Futures on crude oil and natural gas allow them to lock in prices for future delivery, mitigating the risk of cost variations during transit. Another instrument is call and put options, which are useful for hedging against volatility in transportation costs. Call and put options are also useful for hedging against volatility in transportation costs, with call options securing future buying prices and put options protecting against declining selling prices. Unlike standardized futures, forward contracts can be tailored to hedge specific transportation-related risks through agreements between two parties. Swap agreements, including commodity swaps and interest rate swaps, are well-known tools. Commodity swaps stabilize transportation costs by exchanging floating price payments for fixed price payments, while interest rate swaps help manage fluctuations in interest rates when financing transportation infrastructure with debt.

On the insurance side, global insurers such as Allianz, AIG, and Zurich Insurance Group provide marine insurance, including hull and machinery insurance, which reduces financial exposure from accidents by covering physical damage to tankers and vessels during transportation. Cargo insurance offers financial compensation for unforeseen incidents, protecting against the loss or damage of energy products in transit. Political risk insurance mitigates losses from geopolitical instability, expropriation, and other political events. For more complex and larger risks, Lloyd's is the major underwriter.

Trends and challenges. Transporting oil and gas was traditionally considered a low capital risk compared to upstream and downstream activities, as the midstream segment was viewed as a smaller part of the overall supply chain. It relied heavily on a predictable upstream supply and strong demand from downstream commercial, industrial, and retail consumers. Currently, there is a strong push towards renewable energy sources in the sector, with firms focusing on reducing the carbon footprint of energy transportation through innovations in materials and the integration of digital technologies that improve operational efficiency and safety. Ensuring the security of energy infrastructure against

cyberattacks and natural disasters is becoming a higher priority. Additionally, since energy transportation routes often cross multiple countries, control over key chokepoints like the Strait of Hormuz or Malacca is becoming even more critical due to piracy and political instability with the potential to influence global energy markets.

3 Measuring energy transportation uncertainty

The main objective of this section is to describe how to construct an indicator measuring the degree of uncertainty in energy transportation. This will allow us to analyze its evolution over time, understand its determinants, and assess its impact on oil markets and geopolitical risk.

Constructing a Newspaper Index. The methodology for creating the energy transportation uncertainty index follows the approach used by Baker et al. (2016) for constructing search-based indices of economic policy uncertainty. This method has been adapted to recent literature on supply chain disruptions (Young et al., 2021; Burriel et al., 2024), focusing on identifying disruptions or adverse events impacting the *midstream* of energy supply chains by measuring the relative frequency of articles containing specific keywords grouped into three semantic blocks.

(i) **Energy (E)** - This group includes terms such as ”oil”, ”gas”, ”petroleum” and ”energy” as well as instances where the news focuses on the industry of transmission pipelines for transporting crude oil and natural gas.

(ii) **Transportation (T)** - This group contains the terms ”midstream”, ”pipeline”, ”transport”, ”rail”, ”barge”, ”tanker”, ”truck”, ”shipping”, ”storage terminal”, and ”infrastructure”.

(iii) **Uncertainty (U)** - This group consists of words that convey negative sentiment, such as ”threat”, ”warn”, ”fear”, ”risk”, and ”concern”.

For an article to be classified as reflecting a threat to energy transportation, a keyword from the energy-related block must appear in the headline or leading paragraphs and a word from the first (E) and second groups (T) must be present within a range of 5 words of each other. Additionally, the article must have a word count exceeding 150 words. Articles that are summaries of news, corporate digests, calendar of events, rankings-based content, routine general and market/financial news , news agency materials, letters, headline-only content, personal announcements, contract tenders, obituaries, TV/Radio listings are excluded to ensure that only supply-side events are selected. This text-based procedure thus ensures the accurate identification of articles related to disruptions in the midstream of the energy industry. From January 1969 and December 2024, monthly data is gathered from a wide range of English-language media sources *worldwide*, including the US, the UK, Ireland, Canada, Australia, India, Singapore, Japan, New Zealand, Hong Kong, Malaysia, South Korea, Taiwan, China, Indonesia, the Philippines, Africa, Saudi Arabia, Qatar, the UAE, Oman, Russia, Spain, Germany, and France, more about these sources can be seen in Appendix A.

The main events. This index tracks the frequency of media discussions concerning energy and its transportation, serving as an important tool for evaluating their economic impact. Figure 1 illustrates key events that have caused major fluctuations in the index, thereby tracing its evolution and validating its effectiveness in identifying threats to energy transportation. The analysis involves reviewing articles that match specified search criteria on specific dates, which helps accurately pinpoint the factors influencing changes in the index. The index effectively tracks events and dynamics distinct from those triggered by supply disruptions Burriel et al. (2024) and geopolitical tensions Caldara and Iacoviello (2022), providing a unique perspective in the energy literature.

The sabotage of a vital Iranian oil pipeline by Iraq marked the onset of events that led to the Iran-Iraq War (1980-1988), significantly impacting global oil markets. This conflict

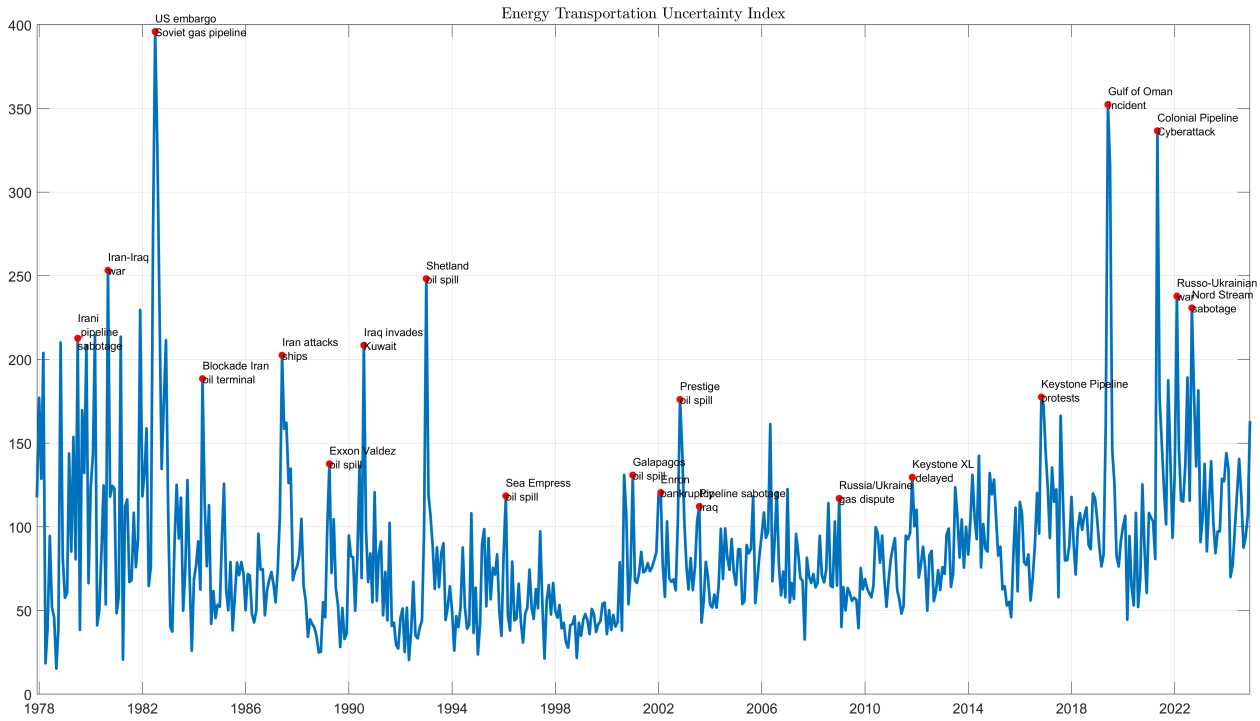


Figure 1: The Energy Transportation Uncertainty Index 1978-2024

targeted critical oil infrastructure in both countries. The resulting sabotage and oil terminal blockades severely limited Iran and Iraq's ability to export oil. Despite this, the impact on the global oil market was asymmetric. Many oil-exporting countries not directly involved in the conflict benefited from higher prices and seized the opportunity to increase their market share. Some of these countries increased production to partially compensate for the supply shortfall caused by the conflict, thereby mitigating some of the upward pressure on oil prices.

In the early 1980s, the United States imposed an embargo on equipment intended for the construction of a Soviet gas pipeline to Western Europe. This move was designed to cut potential Soviet foreign exchange earnings and hinder the development of energy infrastructure that would increase Europe's dependence on Soviet gas, thereby weakening the Soviet economy and geopolitical influence. However, it also created friction in US-European relations, as European countries sought to diversify their energy sources and secure long-term gas supplies.

In 1984, Iraq initiated the *Tanker War* by attacking Iran's major oil shipment ports and oil tankers. Iraq's strategy was to pressure Iran into retaliating by closing the Strait of Hormuz to all maritime traffic, thereby drawing U.S. intervention. However, the U.S. only intervened later in the conflict to protect allied Kuwaiti tankers. Interestingly, this event is not captured by Caldara and Iacoviello (2022) index despite the Tanker war being a major geopolitical event in the 80's.

After a prolonged stalemate at sea, despite heavy losses on land, the Tanker War reignited in May 1987 with an Iraqi missile attack on the USS Stark. The number of attacks doubled compared to 1986. In response, the Reagan administration developed a plan to escort tankers, following discussions with Kuwait. A significant incident occurred during the first U.S.-escorted convoy when the supertanker SS Bridgeton struck an Iranian mine in July. In the following month, a Texaco supertanker loaded with Iranian oil also struck a mine. Although the U.S. escorting campaign mitigated some risks associated with oil shipping, the Iranian Prime Minister warned that the Persian Gulf would remain heavily mined and continue to be a dangerous region for shipping.

The war concluded in 1988, yet tranquility in energy transportation did not follow. In the spring of 1989, the Exxon Valdez oil spill occurred, becoming a major environmental disaster and making worldwide headlines as one of the most significant ecological catastrophes in US history. The period from the Exxon Valdez incident through the early 2000s was characterized by several events predominantly involving oil spills. A major exception occurred in August when Iraq invaded Kuwait, bringing uncertainty to the Gulf region. In response to the coalition's military actions in Iraq and Kuwait, Iraqi forces engaged in environmental terrorism by dumping oil into the Persian Gulf, resulting in the massive Gulf War oil spill.

In 1993, another significant oil spill occurred when a tanker ran aground off the Shetland Islands, Scotland, contaminating the local marine ecosystem and causing considerable economic losses in the fishing, aquaculture, and tourism industries. These incidents

significantly shaped public perception and spurred policy discussions on oil transportation safety, leading to increased compliance costs for tanker owners and the insurance industry due to intensified regulatory scrutiny and the adoption of stricter safety standards. The following period was relatively calm, only disturbed by the Sea Empress oil spill. The year 2001 started with another environmental disaster, with oil spilled along the Galapagos. The next peak event was the Enron scandal, which led to the subsequent bankruptcy of the firm. Enron owned a vast portfolio of oil and LNG transportation, shipping, freight, and risk management operations.

The last major oil spill flagged by the index was the Prestige oil spill. This incident, involving a single-hulled oil tanker, stands as the largest environmental disaster in the history of both Portugal and Spain. The degree of contamination exceeded that of the Exxon Valdez due to the larger amount of oil spilled, and the toxicity was considered higher because of elevated water temperatures. The situation was exacerbated by mismanagement from the French, Spanish, and Portuguese governments, who refused to allow the ship to dock. This incident led to the banning of single-hull oil tankers from European ports.

In 2003, the Iraq War began, introducing potential uncertainties, though not to an extreme extent. The major incidents included sabotage to vital pipelines and numerous oil well fires, which were set in an attempt to hinder attacking air forces.

Europe, which had been largely shielded from energy-related uncertainties for many years, saw instability increase throughout 2008 due to numerous contract and pricing disputes between Russian and Ukrainian gas/transit companies. This escalated into the 2009 Russia-Ukraine gas dispute, during which Russia cut off gas supplies to Ukraine. In the week following the cutoff, all Russian gas flows through Ukraine ceased for 13 days, fully cutting off supplies to Southeastern Europe—a region heavily dependent on Russian gas—and to a lesser extent, other European countries. These events led to major financial losses, with Gazprom estimated to have lost $1.5 billion in revenue due to halted

sales. However, the reputational costs were even more significant, as the crisis negatively and potentially irreversibly impacted the reputations of Russia as an energy supplier and Ukraine as a transit country. This event was merely the beginning of future incidents between these two countries, as subsequent events would demonstrate.

At the end of 2011, the Obama administration postponed the decision on the controversial tar sands Keystone XL pipeline until after the upcoming US elections.

In early February, Ukraine's state-owned oil and gas company Naftogaz sued Chornomornaftogaz, the Crimean oil and gas firm, for 1 billion euros over delayed debt payments. By the end of the month, Russia annexed Crimea and nationalized assets belonging to Ukraine, with no compensation paid.

In January 2015, a pipeline near the Yellowstone River broke, causing an oil spill that leaked environmental hazards into the surrounding area and disrupted the local tap water supply. On 25 November 2015, Gazprom halted its exports of Russian natural gas to Ukraine, with Ukraine claiming it could fulfill its gas supply needs solely from European Union states.

In 2016, protests by indigenous groups and environmentalists erupted in response to the construction of a pipeline extending from the Bakken oil fields in western North Dakota to Illinois. The protests concluded in February 2017 when the President Donald Trump signed an executive order that permitted the construction of the pipeline to continue.

This escalation reached a critical point in 2019, a year marked by a major event in the index. In May 2019, the U.S. withdrew from the Iranian nuclear deal and reimposed sanctions on Iran, citing violations of the agreement. That same month, four oil tankers were sabotaged in the Gulf of Oman. The situation intensified in June with two additional tanker attacks near the Strait of Hormuz. Tensions eventually deescalated following the assassination of General Soleimani and the onset of the pandemic. However, this event marked the end of a period of relative calm in energy transportation that had per-

sisted since the early 1990s. Economically, the uncertainty led to a sharp increase in war risk insurance premiums for the Persian Gulf, with rates for supertankers jumping from US$50,000 to US$185,000.

Interestingly, the main events during the pandemic in 2020 were not necessarily related to the pandemic itself. The first was a major oil spill in a North Carolina nature reserve, caused by a crack in the pipeline. This incident led to the largest U.S. gasoline pipeline spill on record and triggered a massive cleanup and repair operation involving over 200 workers. Later in the same year, in October, the U.S. sold Iranian fuel that had been seized en route to Venezuela.

In May 2021, a new threat to energy transportation emerged with a ransomware attack on the Colonial Pipeline, the largest fuel pipeline system in the United States. This incident marked the largest cyberattack on oil infrastructure in U.S. history. The attack halted pipeline operations, leading to fuel shortages in several states.

In 2022, Russia invaded Ukraine, prompting Ukraine to halt natural gas exports.

Shortly after the invasion began, sanctions were imposed, leading to a Russia-EU gas dispute as payments for pipeline natural gas were switched to rubles. Later in the year, the Nord Stream pipeline was sabotaged, resulting in major leaks and extensive damage in the pipeline. Although this sabotage had no immediate effect on European energy supplies—since Nord Stream 2 never entered service and Nord Stream 1 had been shut down since August—it caused natural gas prices to surge due to fears of further energy disruptions and potential sabotage in the Baltic Sea.

In January 2024, the increasing number of attacks on commercial vessels in the Red Sea by Yemeni rebels, in response to the Israel-Hamas conflict, presented tanker operators with difficult decisions. Some tankers chose to navigate around Africa, and this was particularly evident among Qatari liquefied natural gas carriers that decided to temporarily avoid the Suez Canal. Supertankers were less affected as they generally avoid the canal due to their size.

In the following section, the time series will be implemented with the events described above.

Comparison with existing uncertainty indices A correlation analysis between the ETU index and known uncertainty indices helps place this new ETU measure in the wider uncertainty literature. The analysis includes the EPU index by Baker et al. (2016), the OPU index by Abiad and Qureshi (2023), and the TPU index by Caldara et al. (2020). The ETU index exhibits low correlations with all three established measures, with each correlation falling below 0.26. This finding provides robust support for the idea that the ETU index measures a distinct economic phenomenon. Current uncertainty indices do not capture this well. The weak link with EPU (ρ = 0.25) shows that general economic policy uncertainty sometimes aligns with energy transportation disruptions. These events are mostly distinct, each with its own causes. The weak correlation with OPU (ρ = 0.21) shows that uncertainty in oil prices and energy transportation systems represent different aspects of energy market dynamics. Oil price uncertainty primarily relates to the balance between demand and supply. In contrast, energy transportation uncertainty focuses on infrastructure and logistical issues in the midstream sector. The correlation with TPU (ρ = 0.22) shows that energy transportation uncertainty is a specific risk. This risk is different from general trade policy issues. These findings prove that we need a specific measure for uncertainties in the midstream energy sector. The ETU index plays an important role in filling a gap in uncertainty measurement research. The low correlations across all indices show that the ETU index provides novel information not contained in existing measures.

4 Methodology and data overview

In this section, the study empirically explore the effects of uncertainty surrounding news about energy transportation. To achieve this, a VAR model is employed with monthly data from January 1974, marking the structural changes due to the rising influence of

governments in oil-producing countries, to December 2023. This dataset spans 600 observations and includes real oil prices, world oil production, world oil inventories, world industrial production, geopolitical risk, and real fertilizer prices.

4.1 Empirical Framework

In this study, a VAR model of order p that integrates n endogenous variables is used. The model is formulated as shown in (4.1):

$$y_t = c + \sum_{i=1}^{p} \mathcal{A}_i y_{t-i} + \varepsilon_t \tag{4.1}$$

where, y_t represents an $n \times 1$ vector of endogenous variables in time period (month) t. The right-hand side includes an $n \times 1$ vector of constants, and $i = 1, \ldots, p$ time lags of the vector of endogenous variables, y_{t-i}, with corresponding $n \times n$ reduced form coefficients matrices $\mathcal{A}_i$. As it is standard in the literature on oil market VARs (see Kilian and Zhou (2022)) with monthly data, we set the lag length of the VAR to $p = 24$, in the robustness check this choice is tested. Finally, ε_t is an $n \times 1$ vector of reduced-form innovations, which are assumed to be independently and normally distributed with zero mean and positive definite covariance matrix, Σ. Having established the foundational framework for this analysis, the paper now proceeds to outline the dataset. This includes detailing the endogenous variables involved, the sources of our data, and the econometric approach employed in the identification strategy.

Data. The long tradition of modeling the global crude oil market has led to a consensus on a set of variables rich enough to adequately capture the effects of oil-related shocks, as outlined by Kilian and Zhou (2022). The vector of endogenous variables in Equation 4.1 is of dimension 6×1 and given by $y_t \equiv \left(p_t^o, \Delta q_t^w, \Delta i_t^w, y_t^w, \xi_t, p_t^f\right)'$. First, real oil prices are measured by the US refiners' acquisition cost of imported crude oil, deflated by the

headline US CPI. Second, the producer side is gauged by the growth in world oil production from the US Energy Information Administration (EIA). Third, changes in global crude oil inventories, serving as a proxy for inventory demand, are calculated based on OECD petroleum stocks (EIA1976) and U.S. crude oil and petroleum stocks (EIA1533, EIA1541), using the methodology developed by Kilian and Murphy (2014). Fourth, world industrial production is used as a proxy for global demand. This metric comprises the industrial production of OECD countries plus six additional nations: Brazil, China, India, Indonesia, Russia, and South Africa, as outlined in Baumeister and Hamilton (2019). Fifth, geopolitical risk is measured using the news-based measure of adverse geopolitical events and associated risks created by Caldara and Iacoviello (2022). Sixth, an additional commodity price used as an anchor variable, as described in Boer et al. (2023), is included. This variable is the real fertilizer price index by the World Bank, which includes prices of DAP, phosphate rock, potash, TSP, and urea, deflated by the headline US CPI. While this variable is influenced by the aggregate commodity demand, it remains unaffected by energy-specific demand shocks on impact. Figure 2 illustrates the time series data of the variables used in the baseline VAR over the sample period. Finally, the dataset begins in January 1974, a pivotal time marked by a structural change due to the rising influence of governments in oil-producing countries. For further source details, see Table 2.

The identification The reduced form VAR presented in Equation 4.1 can be rewritten in the structural form as:

$$\mathcal{D}y_t = c^* + \sum_{i=1}^{p} \mathcal{A}_i^* y_{t-i} + \eta_t \tag{4.2}$$

where $c^* = A_0 c$, $A_i^* = A_0 \mathcal{A}_i$, and $\eta_t \sim \mathcal{N}(0, I)$. The reduced-form innovations, ε_t, can be expressed as linear combinations of the structural shocks η_t: $\varepsilon_t = A_0^{-1}\eta_t$. For the structural shocks η_t to be economically meaningful, they must be subjected to identifying restrictions.

While the ETU index follows a plausible narrative a few points shall be noted con-

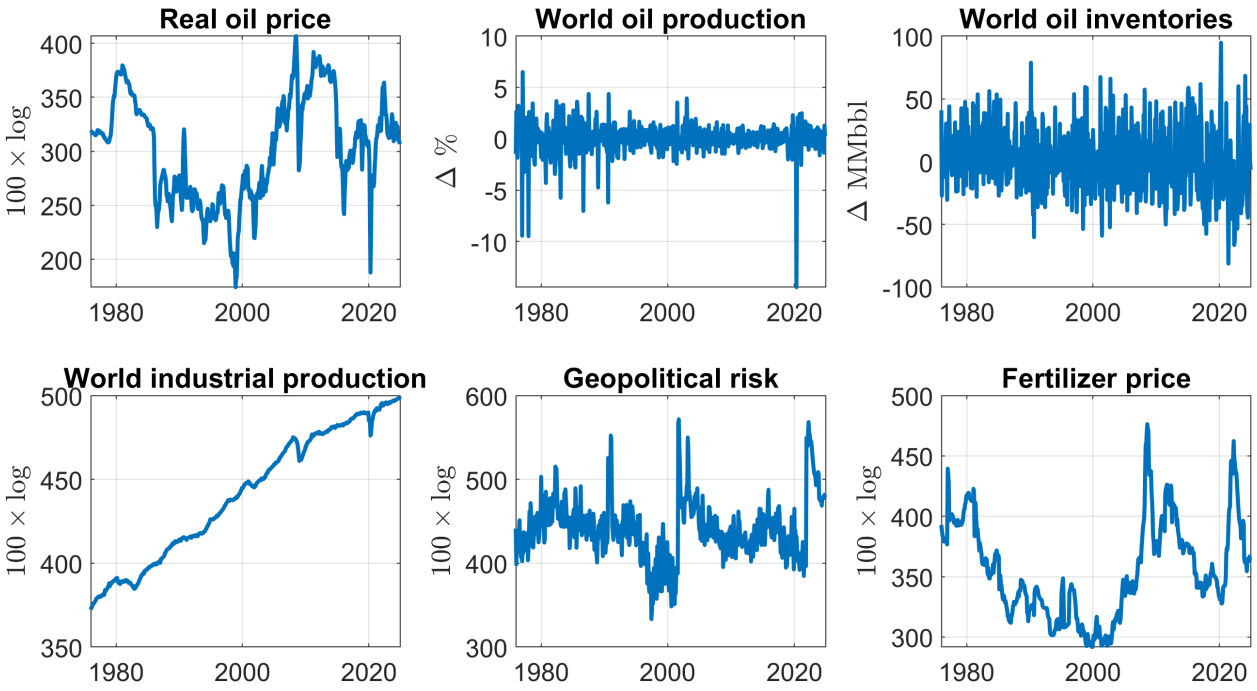


Figure 2: The data series in the baseline model

cerning identification. First, the ETU index might capture not only exogenous transportation uncertainty shocks but also endogenous responses to other economic developments. News coverage itself can be influenced by prevailing economic conditions, potentially introducing reverse causality concerns where oil price movements might influence reporting on transportation uncertainty. Second, despite the rich set of variables in the model, omitted factors could simultaneously affect both the ETU index and our outcome variables of interest. For example, broader geopolitical developments could drive both transportation disruptions and oil price movements through channels not fully captured in the framework. Nevertheless, the historical validation of the ETU index in Section 3 through its alignment with well-documented exogenous events (such as the Nord Stream sabotage and the Colonial Pipeline cyberattack) provides confidence that the identification strategy does capture meaningful exogenous variation in transportation uncertainty. The robustness checks in Section 6 support these findings. They exclude recent pandemic and war periods and use different ordering specifications. This helps us address some of these methodological challenges.

For this analysis, an identification approach is adopted that imposes sign[1] and correlation restrictions on the shocks, as proposed by Ludvigson et al. (2021) and implemented recently in the energy markets context in Morão (2024). Mathematically, this approach is defined by the expression $\rho_1 = \text{corr}(S_t, \eta_{1,t}) \geq \bar{c}$, where ρ_1 denotes the correlation between S_t, (the index documented in Section 3, which is an external (proxy) variable to the SVAR model), and $\eta_{1,t}$ (the uncertainty shock). This identification method ensures that the correlation between the structural shock η_t and the external variable S_t is always above a certain predefined level ($\bar{c}$). This approach is crucial for capturing the dynamics between unexpected spikes in transportation uncertainty and the structural responses within the SVAR framework.

Estimation The model employs conventional Bayesian methods for estimation, postulating a Minnesota prior with standard shrinkage parameters (Giannone et al., 2015), along with a sum-of-coefficients prior (Doan et al. (1984)) and a dummy-initial-observation prior (Sims (1993)). The hyperparameters are calibrated as follows: overall tightness (λ_1) at 0.25, cross-variable specific variance weighting (λ_2) at 0.35, and lag decay (λ_3) at 1. The variance for the priors on the exogenous variables (λ_5) is set to 10^5. The settings for the sum-of-coefficients tightness (λ_6) and dummy initial observation tightness (λ_7) are 0.1 and 0.01, respectively. This prior on the VAR slope coefficients (β) is modeled as a normal distribution with a zero mean (β_0) and diagonal matrix Ω_0, approximated using the residual variance from an AR(1) regression. The prior for the covariance matrix (Σ) follows an inverse-Wishart distribution with scale matrix S_0 and α_0 prior degrees of freedom. Consequently, the estimated model parameters are biased towards a multivariate random walk, which is a plausible assumption considering the non-stationary dynamics of the variables in question (refer to Figure 2). However, the results remain stable even when we opt for prior values less than one for the AR(1) parameters. The joint posterior

[1]Sign restrictions come from early results using recursive identification and economic theory.

distribution of β and Σ given y does not have an analytical form. However, the conditional posterior distributions for β and Σ are analytically tractable. This allows the use of Gibbs sampling for posterior inference. As per Dieppe et al. (2016), the Gibbs sampling procedure in Bayesian VARs with independent normal inverse-Wishart priors is well-detailed. The algorithm runs for 4000 iterations, including a burn-in period of 3000 iterations. Hyperparameters are set based on common values found in the literature, as detailed in Table 3. The identification strategy used in this study, along with a detailed description of the dataset used, will be discussed in the next subsection.

5 The impact of energy transportation uncertainty

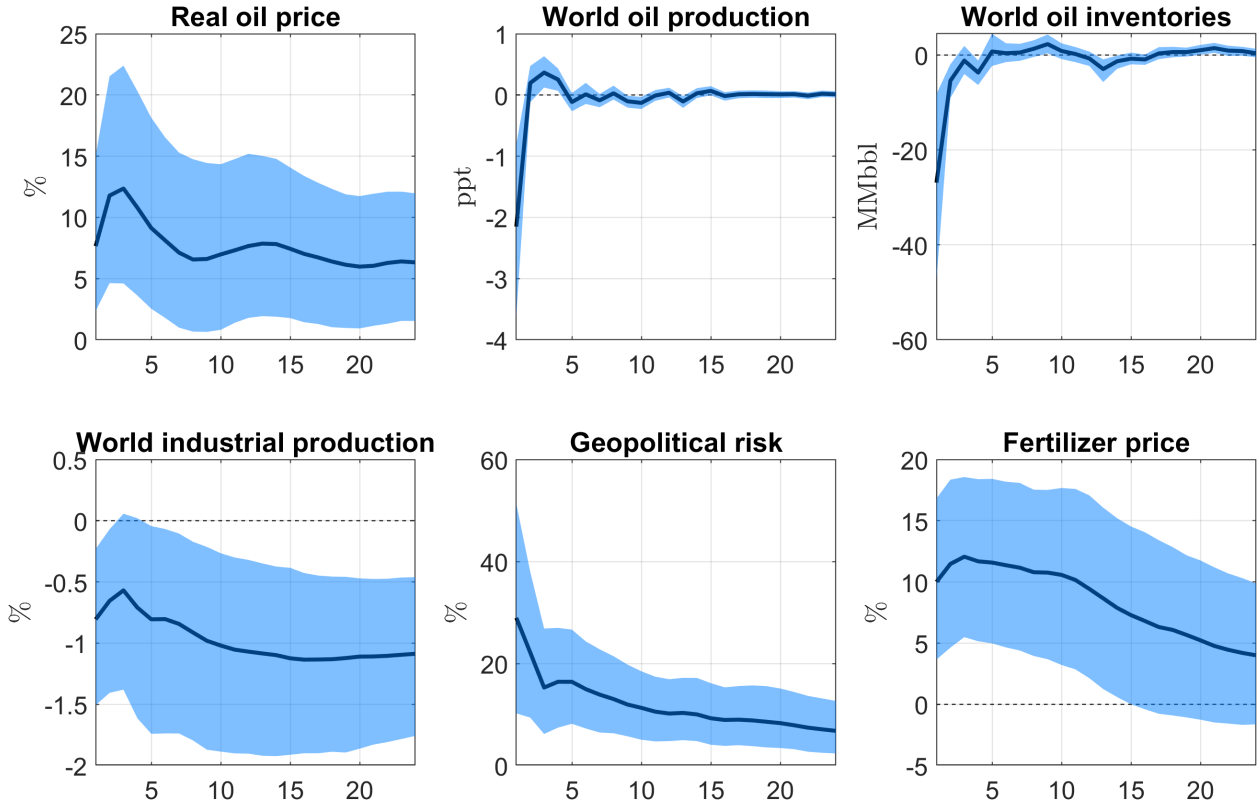
To assess the effect of energy transportation uncertainty on the model variables. This analysis will be conducted in three different parts, enabling effective generation and interpretation of results in this section. The Impulse Response Functions (IRFs) describe how the variables react to the uncertainty shocks. The Forecast Error Variance Decomposition (FEVD) will verify if the shocks are significant sources of fluctuations, and the Historical Decomposition will analyze their role in the historical dynamics of oil prices and geopolitical risk.

5.1 IRF analysis

First, we will examine the IRFs. Figure 3 shows the effects of the uncertainty shock. The objective is to understand how a shock that raises transportation uncertainty affects the model variables. This analysis will provide proper quantification, a clearer understanding of the implications, and a significant contribution to the broader economic analysis of these shocks.

The increase in threats to energy transportation, including geopolitical tensions, conflicts, and infrastructure disruptions, exerts substantial and wide-ranging impacts on the

oil markets and the global economy via several mechanisms.



Note: Impulse responses to a energy transportation uncertainty shocks normalized to increase the real oil prices by 10% on impact. The solid black line represents the (point) posterior median estimate and the shaded areas 68% credible bands.

Figure 3: Impulse responses the shock

First, this shock causes a sudden and significant increase in the real price of oil as consumers compete for a limited supply.

In response to the perceived increase in risk, world oil production decreases significantly on impact; however, it readjusts quickly, returning to its pre-shock level. Initially, world oil production growth decreases significantly in response to perceived increased risk, but it quickly adjusts, returning to its pre-shock rate. As oil production falls, oil inventories are depleted in an effort to smooth consumption.

The shock propagates through the global economy as heightened geopolitical tensions, as measured by the geopolitical risk index, lead to a significant and persistent decline in world industrial production. This result validates the index's effectiveness

in assessing geopolitical risk accurately, as rising uncertainty dampening output through the channels of investor and consumer confidence is a well-established result in economic literature.

Finally, the persistent increase in oil prices following the shock also leads to a sustained rise in the price of fertilizer. Fertilizer, crucial for crop production, demands substantial energy for its manufacturing process. Therefore, these findings demonstrate that the agricultural sector faces direct consequences from transportation uncertainty because of its dependence on energy prices.

The increase in fertilizer prices puts pressure on farmers' and food producers' profits, triggering a decline in agricultural output, impacting food security and prices, and potentially sparking social unrest. The substantial increase in the real oil price, coupled with reductions in oil production and declining inventories, aligns with the interpretation of a *flow supply shock*.

5.2 Historical importance

Variance decomposition analysis. After analyzing shock transmission in the structural model, we can now tackle key issues from the existing literature concerning the drivers of fluctuations in the real price, production, and inventories of oil. To do this, the study employs a Forecast Error Variance Decomposition (FEVD). Table 1 presents the percentage contribution of the uncertainty shocks to the variance of each specific variable.

In the immediate aftermath $(h = 1)$, the uncertainty shock explains a significant share of the variation across all variables in the study.

The relatively stable contribution of transportation uncertainty to the variance in real oil prices (around 13%) indicating that while transportation logistics are important to oil prices, other factors such as OPEC decisions and global demand play more dominant roles. This stable response suggests that market participants are able to quickly adjust their expectations and hedging strategies in response to the shock impact.

Table 1: Forecast error variance decomposition

h	p_t^o	Δq_t^w	Δi_t^w	y_t^w	ξ_t	p_t^f
1	12.84	26.76	16.83	15.59	21.90	24.57
6	14.74	24.62	16.74	9.97	25.17	29.23
12	13.21	24.53	16.88	12.17	26.87	25.33
24	13.67	25.43	17.00	18.81	28.25	19.35

Notes: The table displays the median values of the forecast error variance decomposition for the shock results at the 1-month, 6-month, 12-month and 24-month horizons.

These shocks account approximately around 25% part of the variance in oil production changes, particularly in the short run. This high sensitivity is mainly due to immediate disruptions in supply chains that directly impact the pace and scale of production adjustments. Over time, the impact of these shocks decreases, indicating that oil producers are likely adapting by diversifying transportation routes and increasing storage capacity to mitigate the effects of future disruptions on production changes.

The consistent contribution on inventories (around 16%) underscores the role of inventories as a strategic buffer. This persistence across time frames suggests that inventory management practices are closely tied to anticipations and reactions to transportation uncertainties, reflecting a proactive approach to risk management in the oil sector.

The rising impact of transportation uncertainties on global industrial production, which increases from 14% to 18% over time, reveals a nuanced economic dynamic. Initially, industries display some resilience by employing contingency plans and existing inventories to handle disruptions in energy transportation. However, as time progresses and these uncertainties persist, their effects become more pronounced, exposing vulnerabilities within supply chains and energy dependencies. As adaptive capacities diminish and the costs of alternative logistics solutions escalate, significant disruptions in manufacturing processes and delivery schedules will eventually occur.

The slight increase in the contribution of transportation uncertainty to geopolitical

risk over time (from 20.39% to 22.37%) suggests a feedback loop where transportation disruptions enhance geopolitical tensions, which in turn lead in geopolitically sensitive regions to more uncertainties in transportation.

The contribution on fertilizer prices peaks at 6 months (27.64%), which then declines, reflects a dependency of fertilizer prices on natural gas and petroleum-based products, which are influenced by transportation conditions. The initial high impact represent immediate price adjustments due to fears of supply shortages, which stabilize over time as the market adjusts to new realities.

Historical decomposition analysis. The main objective is to explore the role of transportation uncertainty in oil prices and geopolitical risk, and to analyze its influence during significant historical episodes. The historical decompositions in Figure 4 provide additional information on how the shock cumulatively affected the evolution of oil prices and geopolitical risk. The decomposition's orange bars show the exogenous component arising in response to the uncertainty shocks, i.e., the contribution to historical oil prices and geopolitical risk that correlates with energy transportation news shocks. The orange bars represent the contribution arising from OPU shocks to geopolitical risk. The black line shows the actual geopolitical risk data in percentage deviations.

Energy transportation uncertainty shocks significantly impacted oil prices and geopolitical risk during three main episodes. Examining specific episodes provides valuable insights into the timing and context of these fluctuations and the associated geopolitical risks.

The first major period began with the U.S. sanctions on the Soviet pipeline in Europe and extended to the Tanker War. This era was characterized by intense geopolitical uncertainty, significantly impacting the global energy market. It was a long episode of high uncertainty, not just due to disruptions in key transit routes which increased the risk premium on oil, driving up prices, but also due to intensified potential geopolitical conflicts

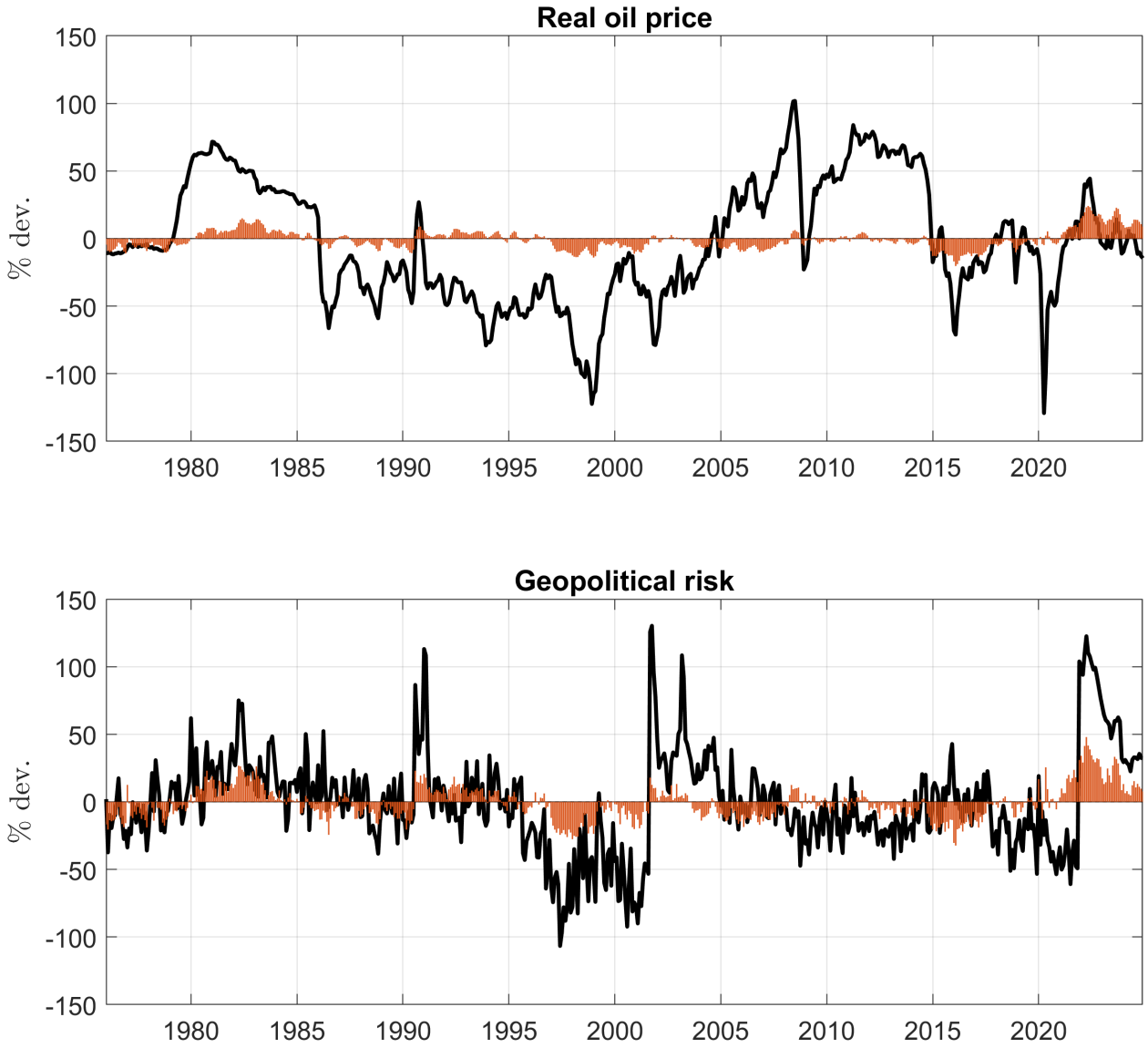


Figure 4: Historical decomposition of oil prices and geopolitical risk

in the region. Investors and traders demanded higher returns to compensate for the perceived risk, pressuring oil prices despite new incoming oil production in Alaska and the North Sea fields.

In contrast, the period from the mid-1990s to the early 2000s was relatively calm. During this period, and particularly by the Asian crisis of 1997–1998, both oil prices and geopolitical risk were at historically low levels. The negative spike can be partially explained by historically low levels of energy transportation uncertainty. Reduced uncertainty significantly contributed to this stability, playing a crucial role in maintaining low levels. The predictability of energy supply chains reduced oil market anxiety that plagued the previous two decades, effectively lowering the risk premium on oil transportation, such as tanker insurance, and providing a sense of security among market participants. The relative calm in geopolitical affairs reduced speculative behavior in oil markets as traders did not need to factor in significant risk premiums, further contributing to price stability. Lower geopolitical tensions also meant that oil-importing countries could focus on diversifying their energy sources and investing in efficiency and environmental sustainability without the immediate pressure of high oil prices.

Interestingly, news about transportation uncertainty did not significantly contribute to the surge in the real price of oil between 2003 and 2008, nor can it account for the significant fall after the global financial crisis. This is an important point for index validation purposes, as economic literature, such as Kilian (2009), mainly attributes this increase to higher global demand and not to supply-specific factors in the energy markets.

However, this changed in more recent years when energy transportation uncertainty began to play a significant role. Recent episodes can be attributed to threats to energy transportation, mainly due to the 2022 Russian invasion of Ukraine, which threatened many pipelines between Russia and Europe due to sanctions and sabotage. Consequently, the demand for LNG shipping worldwide dramatically increased. Additionally, rising tensions in the Middle East caused the insurance costs for this crucial commodity to in-

crease significantly.

Evolving nature of energy transportation uncertainty. The evolution of energy transportation uncertainty reflects the changing global energy landscape, technological advancements, and shifting geopolitical dynamics. During the 1980s, transportation uncertainty was predominantly characterized by conventional physical threats. The Tanker War and pipeline conflicts caused major disruptions to energy infrastructure, with geopolitical tensions often resulting in military or political threats to transportation assets. The oil risk premium during this period reflected concerns about physical supply disruptions and the possibility of bottlenecks in some key maritime chokepoints.

The calm period between the mid-1990s and early 2000s came with stable transport systems, diverse supply routes, and improved geopolitical conditions. This era had major investments in energy infrastructure around the world. These investments created backups that lowered the risk of local disruptions. The reduced contribution of transportation uncertainty to oil prices during this period reflects higher resilience within global energy supply chains.

The nature of energy transportation risks underwent a notable transformation beginning in the 2010s. The analysis reveals an increasing contribution to both oil prices and geopolitical risk, but it identifies qualitatively different types of threats. The Colonial Pipeline ransomware attack in 2021 shows this shift. It highlights a cyber vulnerability instead of a traditional physical threat. This incident showed that digital technologies improve efficiency but also bring new disruptions that did not exist before.

Additionally, the growing integration of renewable energy into the global energy mix has begun to alter traditional energy transportation dynamics. Transportation uncertainty in recent years reflects growing concerns about electricity transmission infrastructure, grid stability, and the challenges of intermittency associated with renewable energy sources. As cross-border electricity connections become more important, they introduce

new uncertainties. These uncertainties are unlike those from the fossil fuel-dominated early period in the analysis.

The recent increase in transportation uncertainty, due to the 2022 Russian invasion of Ukraine, also highlights changes in the transportation landscape. This conflict was different. It focused on pipeline infrastructure, energy sanctions, and shifts in trade flows. Previous conflicts primarily threatened maritime chokepoints, but this one highlights new issues. Switching to LNG from pipeline gas creates new geographical risks and transport issues that traditional networks do not have.

These evolving patterns underscore how the economic interpretation of transportation uncertainty shocks must account for the changing nature of energy infrastructure, digital vulnerabilities, and the gradual transition towards more diverse energy sources. While the model treats these shocks as structurally similar across time. Some underlying drivers and transmission mechanisms have evolved significantly, indicating potential non-linearities or time-varying effects that future research could explore in greater detail.

6 Additional results and robustness analysis

Cholesky decomposition. Cholesky decomposition is frequently employed in SVAR models, yet this method has been subject to some debate. Kilian et al. (2022) question the credibility and validity of SVAR models that rely exclusively on recursive identification. Critics argue that these models often lack external validation for their identifying assumptions. Nonetheless, examining different recursive variable orderings continues to be a common practice in empirical macroeconomic research.

Figure 5 contrasts the responses observed in a proxy-identified benchmark with those from a recursively ordered model (illustrated by the green line). The recursively ordered model now exhibits a distinctive hump-shaped pattern in world industrial production. Real oil prices also show a more pronounced hump-shaped pattern; however, the estimates become less certain, as indicated by the wider credible bands. There is also a

more pronounced immediate impact on world oil production, with inventories mirroring the pattern but not the magnitude. Geopolitical risk exhibits a similar pattern but with slightly greater intensity in the months following the shock, although the credibility bands are also wider. Meanwhile, fertilizer prices show a pronounced initial impact, but the effect vanishes in less than six months, unlike the benchmark.

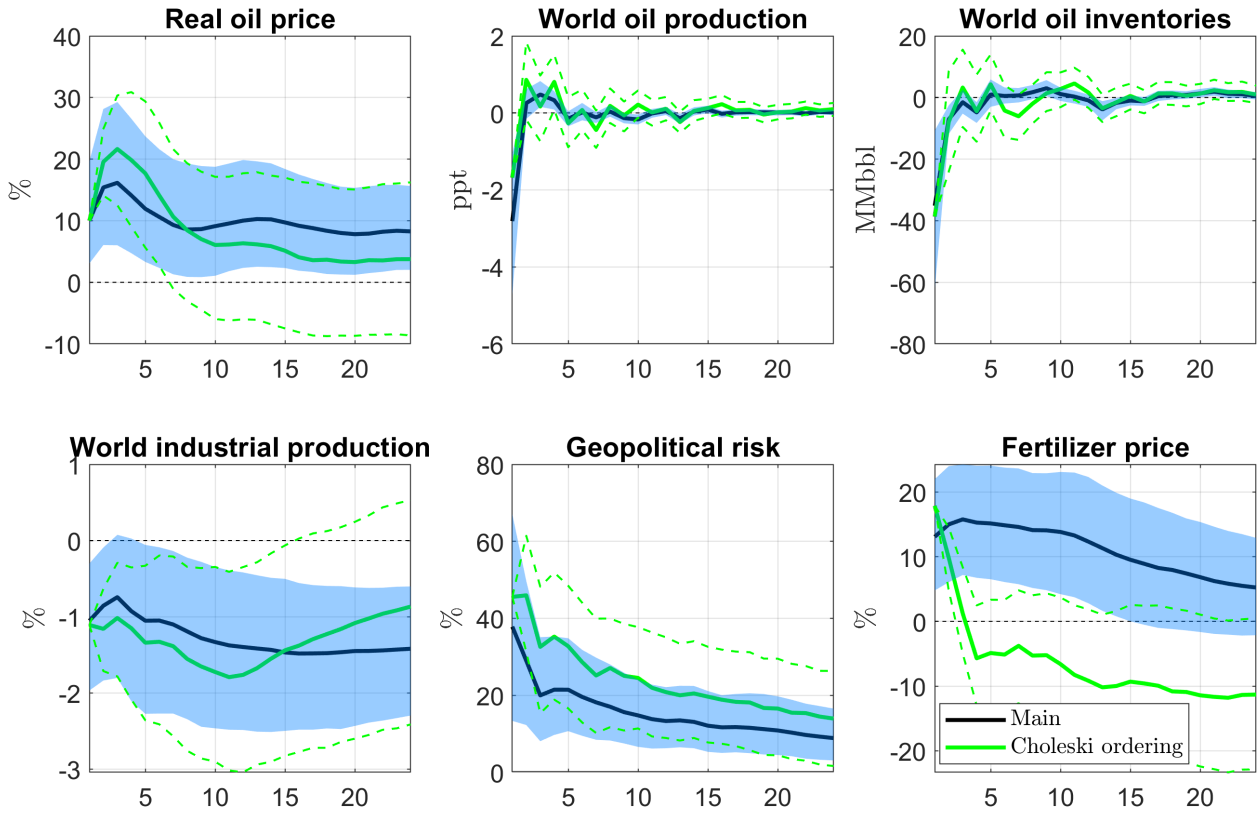


Figure 5: Recursive identification

Removing 2022. Incorporating significant exogenous shocks, such as the 2022 Russian invasion of Ukraine and the pandemic, presents some modeling challenges. The pandemic introduced nonlinearities into the model variables, particularly during the Russia–Saudi Arabia oil price war, which coincided with the COVID-19 pandemic and was marked by anticipated declines in oil consumption. Two months later, Saudi Arabia and Russia agreed to oil production cuts, which resulted in a negative growth spike observed in world oil production. Another nonlinearity is the decline in global industrial produc-

tion, a direct result of the lockdowns.

The conflict between Russia and Ukraine increased geopolitical tensions, leading to energy price volatility. This uncertainty was exemplified by events such as the sabotage of the Nord Stream pipeline, which further destabilized energy supply routes. Additionally, these tensions disrupted supply chains across various sectors, including food and electronics, compounding the economic impact globally.

Appendix Figure 5 contrasts the IRFs from the main model with a scenario excluding the Russian invasion of Ukraine (red line) and all the post-pandemic period (green line). This approach allows us to isolate and understand the latest interactions among changes in energy transportation uncertainty, geopolitical risk, and oil market dynamics. Upon visually inspecting the data, we can draw some relevant conclusions. The similarity in the sign and size of the responses to shocks suggests that recent events—such as the cyberattack on the Colonial Pipeline, the Russia-Ukraine war, and the subsequent bombing of Nord Stream—have not altered the fundamental nature of how energy transportation uncertainty affects the market. This consistency supports the reliability of the benchmark model results, indicating that these events did not introduce significant time-varying effects.

Regional Heterogeneity. While the baseline model captures the global effects of energy transportation uncertainty, significant heterogeneity exists in how these shocks propagate across different economies. Figures 26 to 31 show the responses of Nominal Effective Exchange Rates (NEERs) in six major economies after an ETU shock. The results emphasize that some regions experience more transportation disruptions than others. Japan depends significantly on energy imports and this dependence leads to significant currency depreciation, around 1%, which continues throughout the horizon period. Similarly, the Euro Area also shows a steady drop of about 1%. This decline highlights its weakness to pipeline disruptions, like those seen during the Nord Stream events. In contrast, the

United States, Switzerland and broader Advanced Economies indices show initial appreciation that gradually diminishes over time, consistent with their role as safe-haven destinations during global uncertainty. The UK's response is muted. This response is characterized by minimal exchange rate movements, which likely reflect its balanced energy profile that combines domestic North Sea production with diversified import sources. Additionally, Brexit-related factors may have already integrated energy supply uncertainties into exchange rate expectations. Surprisingly, Emerging economies often experience a rise in their currencies at first. This might seem odd, but it is understandable. Energy-exporting countries enjoy higher expected energy prices and less competition for supply. These advantages can counteract the usual flight-to-quality effects. Energy transportation uncertainty impacts oil-importing and oil-exporting economies in different ways. These effects vary due to differences in energy mix, geography, and financial markets. The exchange rate channel plays a crucial role. It explains how transportation disruptions cause different economic impacts in regions. This supports the global production and price effects noted in the baseline analysis.

Natural gas market. To address the limitations of focusing only on oil markets, the analysis is extended to explore how uncertainty in energy transportation impacts natural gas prices in two major regions. Figure 23 shows clear differences in how natural gas markets react to ETU shocks. These differences come from the unique market structures and transportation systems in place. US natural gas prices rise sharply and quickly, peaking at about 30% around month 5. Then, they gradually decline. This pattern matches the market's strong pipeline network and effective price discovery that reacts fast to transport disruptions. The US market responds quickly due to its hub-based transportation network. When there is uncertainty about pipeline capacity or integrity, regional bottlenecks can quickly drive up prices. This happens without the help of large storage facilities found in international LNG supply chains. The tight confidence bands for the US show

more predictable market behavior. This is because of increased self-sufficiency and better infrastructure redundancy. In stark contrast, Japanese LNG prices display a delayed response pattern, beginning near zero before gradually rising to peak around month 10-12. This lagged pattern shows Japan's distinct market setup. It has long-term contracts with rigid delivery obligations and price formulas that adjust only quarterly or semi-annually, creating substantial lags between transportation disruptions and their price effects. These prices change only after delays caused by transportation disruptions. Japan's wide confidence bands show more uncertainty. This reflects how transportation shocks affect a market with a strong reliance on imports and limited flexibility.

Model specification. To validate the findings from the SVAR analysis presented in this paper, several sensitivity and robustness checks were executed.

The model incorporated exogenous variables, including the OPU index by Abiad and Qureshi (2023) and OPEC news by Känzig (2021) (see Figures 7 and 8). These modifications are intended to improve the understanding of the model's sensitivity to external factors.

Next, different lag lengths were tested, following various criteria to ensure that the model's predictions are not dependent on a specific lag structure. Figures 9, 10, and 11 illustrate this.

The analysis also examined the robustness of conclusions to various prior choices regarding outcomes, as detailed in Figures 12 and 13.

Further investigation explored the model's sensitivity to different hyperparameter settings, as illustrated in Figures 14, 15, and 16. The impact of an alternative assumption in the S_0 matrix was assessed, detailed in Figure 17. Figures 18 and 19 present results without including the priors for unit root and cointegration dynamics.

The effects of including constant and trend terms are examined in Figures 20 and 21, to confirm that the results are not influenced by these factors. Figures 22 and 23 show

alternative measures of world GDP, the index of global real economic activity by Kilian (2009) and the Global Economic Conditions Index by Baumeister et al. (2022). These comprehensive modifications are further elaborated in Appendix B.

Overall, the results are robust, indicating that the model adapts well to changes in specification.

7 Conclusion and policy implications

In addressing recent threats to global energy transportation, understanding the economic repercussions of energy transportation uncertainty in oil markets is crucial. This paper marks a significant advancement in energy literature by employing the text-based algorithm of Baker et al. (2016) to construct a novel index of energy transportation uncertainty. The index, which compiles data from news articles across more than 40 news outlets on a global online news platform, effectively captures major historical events like the Tanker War, the Iran-Iraq War, and the Exxon Valdez oil spill. It also includes recent incidents such as the Nord Stream sabotage, the Colonial Pipeline cyberattack, and the 2019 incident in the Gulf.

To examine the impacts of energy transportation uncertainty, this study employs a canonical oil-market Bayesian SVAR model with shocks related to tensions in energy transportation. These shocks are identified by applying correlation restrictions, using the new index as a proxy.

The study reveals that shocks from energy transportation uncertainty lead to a significant increase in real oil prices as demand outstrips supply due to supply chain problems. World oil production initially decreases due to perceived risks but quickly recovers to pre-shock levels. Despite this quick adjustment in production, oil inventories are drawn down to smooth consumption.

The shock also affects the global economy by increasing geopolitical tensions and causing a significant and persistent reduction in world industrial production. These re-

sults also confirm the index's utility in gauging geopolitical risks, aligning with economic literature that links uncertainty to reduced output through diminishing business and consumer confidence.

To address the effects of energy transportation uncertainty, policymakers should improve the resilience of energy supply chains by diversifying energy sources and strengthening international cooperation on energy security issues. Maintaining strategic reserves and creating flexible regulations to quickly adapt to changes in energy supply emerge as straightforward solutions to reduce the risk of energy shortages and social unrest due to energy supply disruptions. For future research, employing the index to examine the macroeconomic effects across a panel of countries, leveraging the principle that disruptions in energy supply can have widespread economic impacts, is a promising direction for economic research on this topic.

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APPENDIX FOR PUBLICATION

A Appendix: Data and Model

Table 2: Data Description and Sources

Label	Description	Source
p_t^o	real US refiners acquisition cost of imported crude oil	US EIA
Δq_t^w	world oil production	US EIA
Δi_t	global crude oil inventories	US EIA/Kilian and Murphy (2014)
y_t^w	world industrial production	Baumeister and Hamilton (2019)
ξ_t	geopolitical risk historical index	Caldara and Iacoviello (2022)
p_t^f	fertilizer price index	World Bank
opec_t	Oil supply news index	Känzig (2021)
opu_t	Oil price uncertainty index	Abiad and Qureshi (2023)

Notes: Data series are monthly and covering 1974m1-2023m12 period in the main specification.

Table 3: Hyperparameters

	Value	Description	Observation
ρ	1	autoregressive coefficients	Used in all models
λ_1	0.2	overall tightness	$\lambda_1 = 1000$ in model 14 & $\lambda_1 = 1$ in model 15
λ_2	0.9	cross-variable weighting	$\lambda_2 = 2$ used in model 17
λ_3	1	lag decay	Used in all models
λ_4	10^5	exogenous variable tightness	Used in all models

Table 4: Newspaper coverage

News source	Start date	News source	Start date
The New York Times	1980-06	The Japan News	2002-03
The Boston Globe	1987-01	Taipei Times (Taiwan)	2002-06
The Wall Street Journal	1980-01	All Africa	2002-08
New York Post	1997-10	Khaleej Times (UAE)	2011-01
Financial Times	1981-01	Al Jazeera English	2007-07
The Guardian (U.K.)	1982-01	The Moscow Times	1997-02
The Globe and Mail (Canada)	1977-11	St. Louis Post-Dispatch	1988-01
South China Morning Post (HK)	1990-01	The Times (U.K.)	1981-01
The Washington Post	1977-01	Tampa Bay Times	1986-12
USA Today	1987-04	Kyodo News (Japan)	1994-10
The Straits Times (Singapore)	1989-07	Bangkok Post (Thailand)	1989-04
Daily Mail (U.K.)	1995-01	The Salt Lake Tribune	1990-09
The Australian Financial Review	1997-01	The Observer (U.K.)	1981-04
The Daily Telegraph (U.K.)	2000-11	Times of Oman	2009-06
Daily Telegraph (Australia)	1996-07	Arab News	2008-09
Toronto Star	1988-04	Agence France Presse	1991-09
The Independent (U.K.)	1989-01	El País - English Edition (Spain)	2001-01
National Post (Canada)	1990-11	Textline Collection (1980-1994)	1981-01
Irish Times	1993-01	Middle East Economic Digest	1984-06
Daily Herald (Chicago)	1999-01	BBC Middle East	1984-07
The Press and Journal (U.K.)	1998-10	The Jakarta Post (Indonesia)	1994-10
The Economic Times (India)	1997-04	BusinessWorld (Philippines)	1995-09
The Times of India	1997-04	DW English	2006-07
The New Zealand Herald	1997-04	Business Times Singapore	2006-07
Herald-Sun (Australia)	1997-07	New Straits Times (Malaysia)	1995-10
Korea Times (South Korea)	1998-03	China Daily	1993-09

APPENDIX NOT FOR PUBLICATION

B Appendix: Sensitivity analysis

B.1 Robustness: Pandemic and war

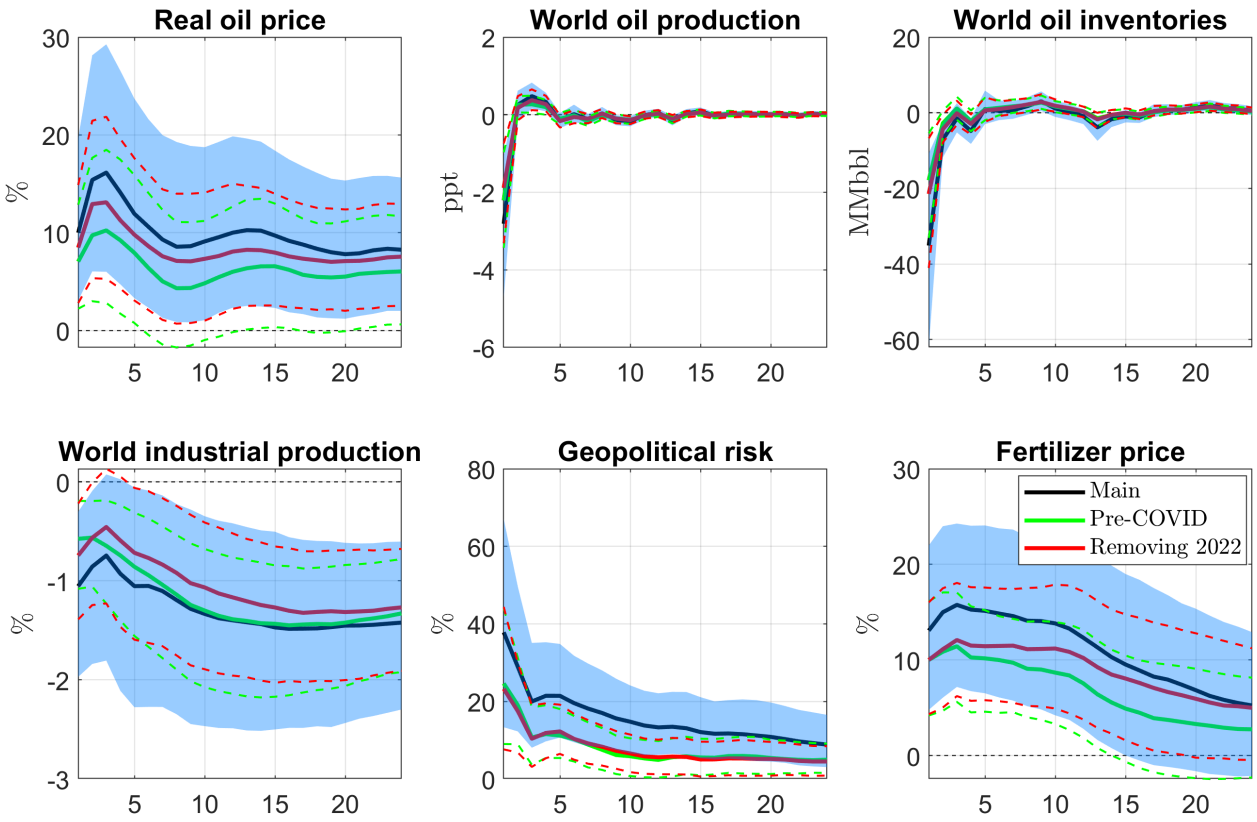


Figure 6: Including pandemic and war

B.2 Robustness: OPU

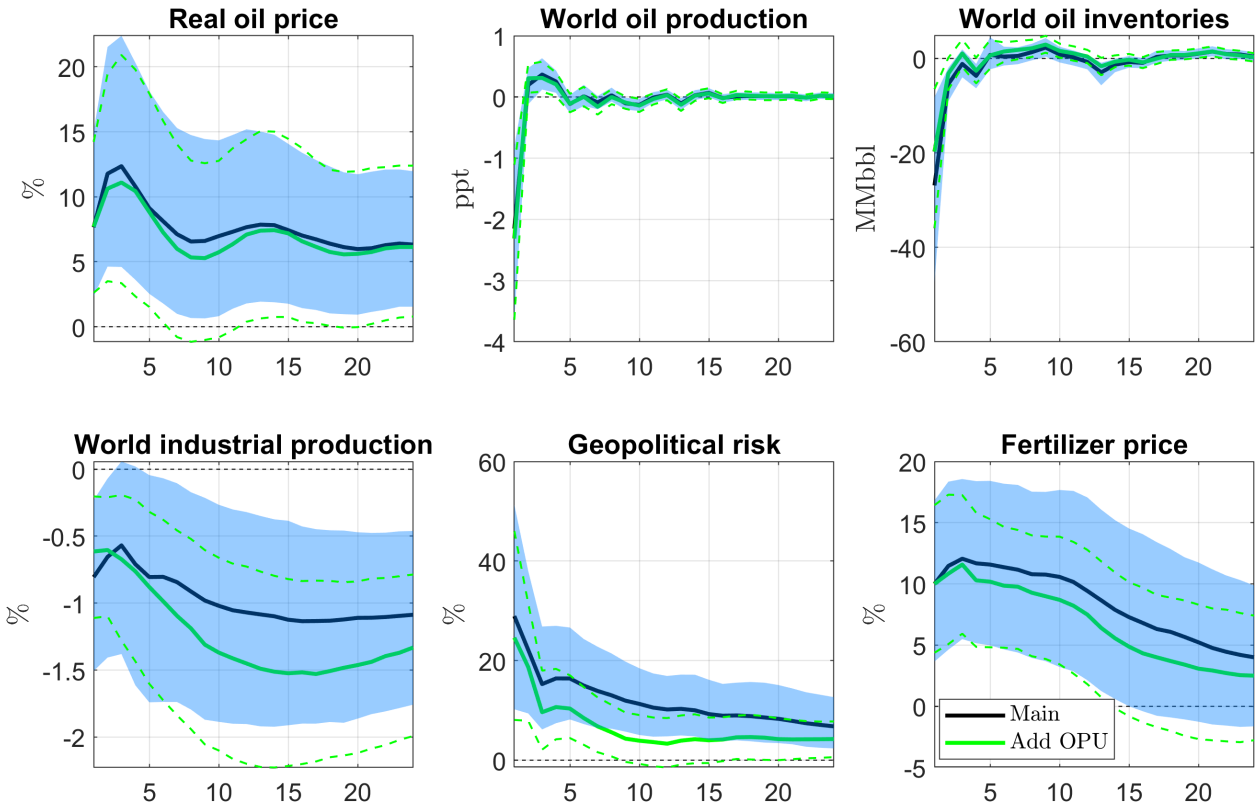


Figure 7: Model including OPU index as exogenous variable

B.3 Robustness: OPEC

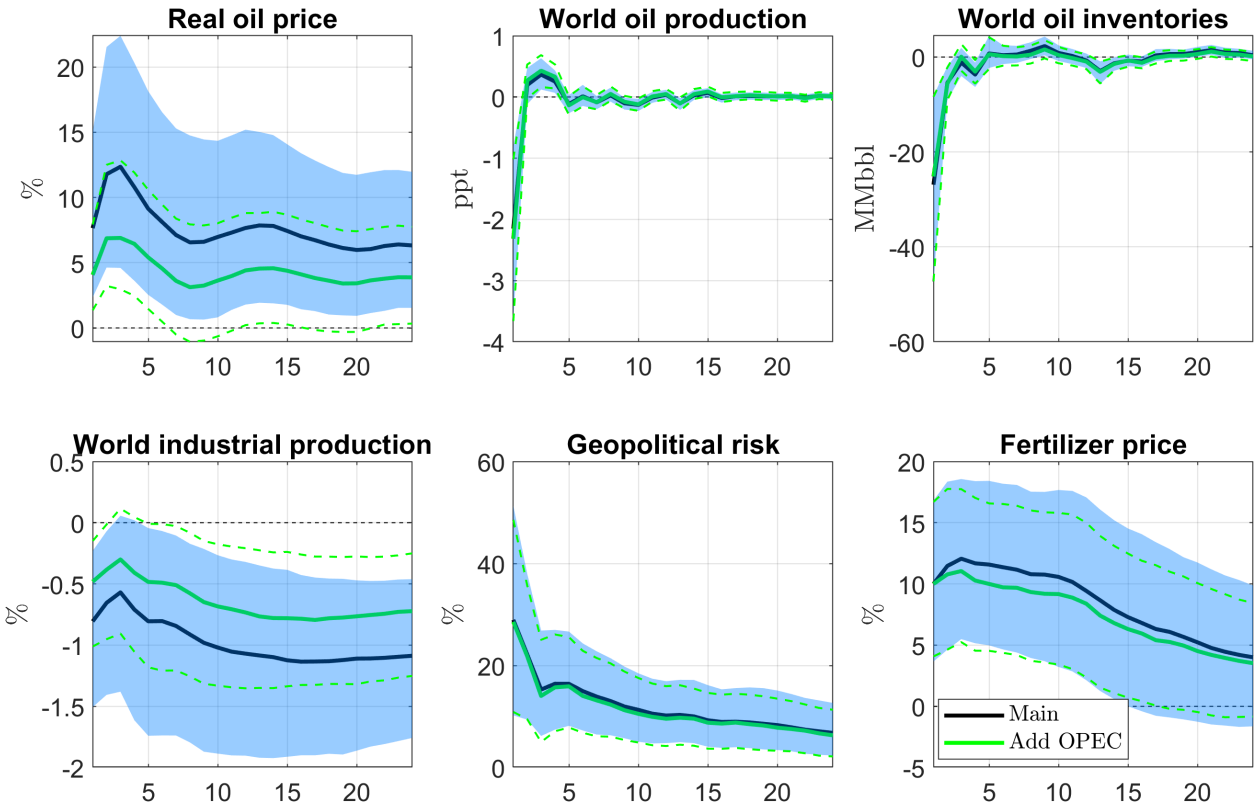


Figure 8: Model including OPEC news as exogenous variable

B.4 Robustness: 3 lag

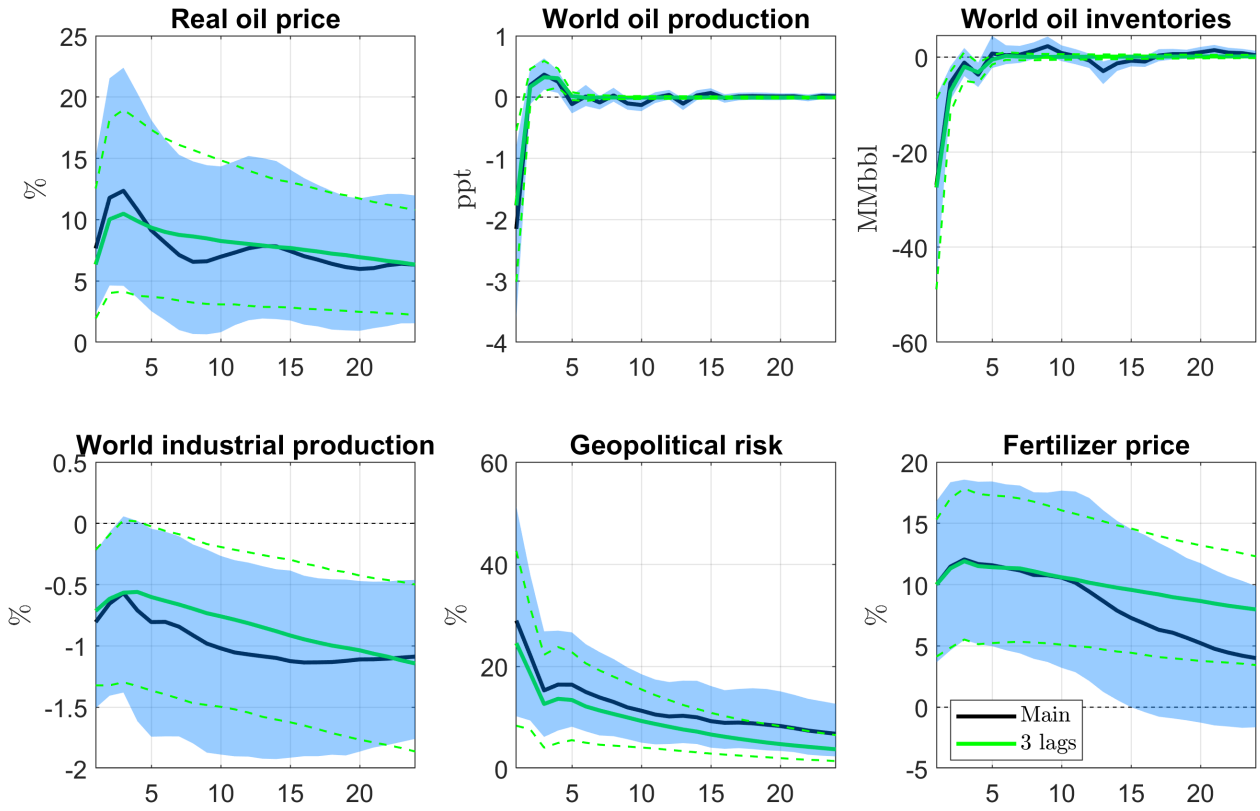


Figure 9: Results from a BVAR(3)

B.5 Robustness: 6 lags

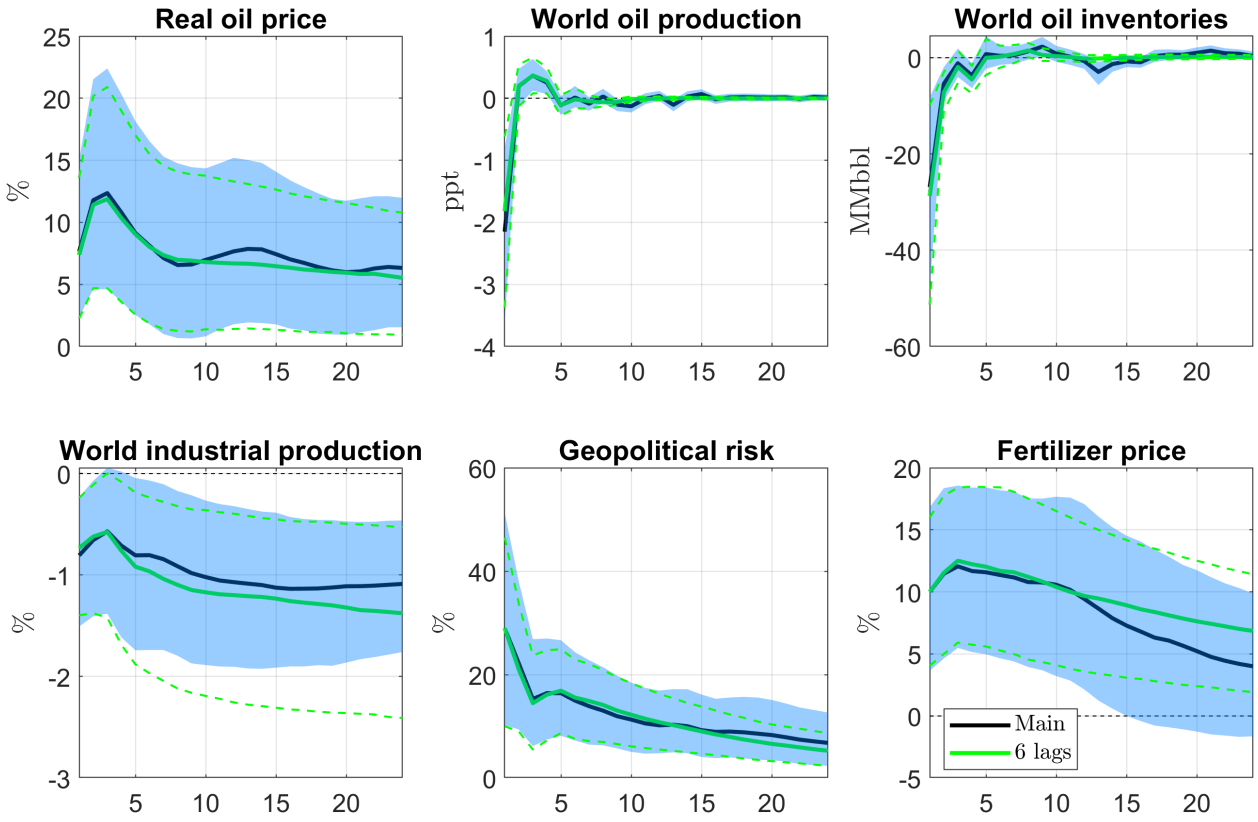


Figure 10: Results from a BVAR(6)

B.6 Robustness: 12 lags

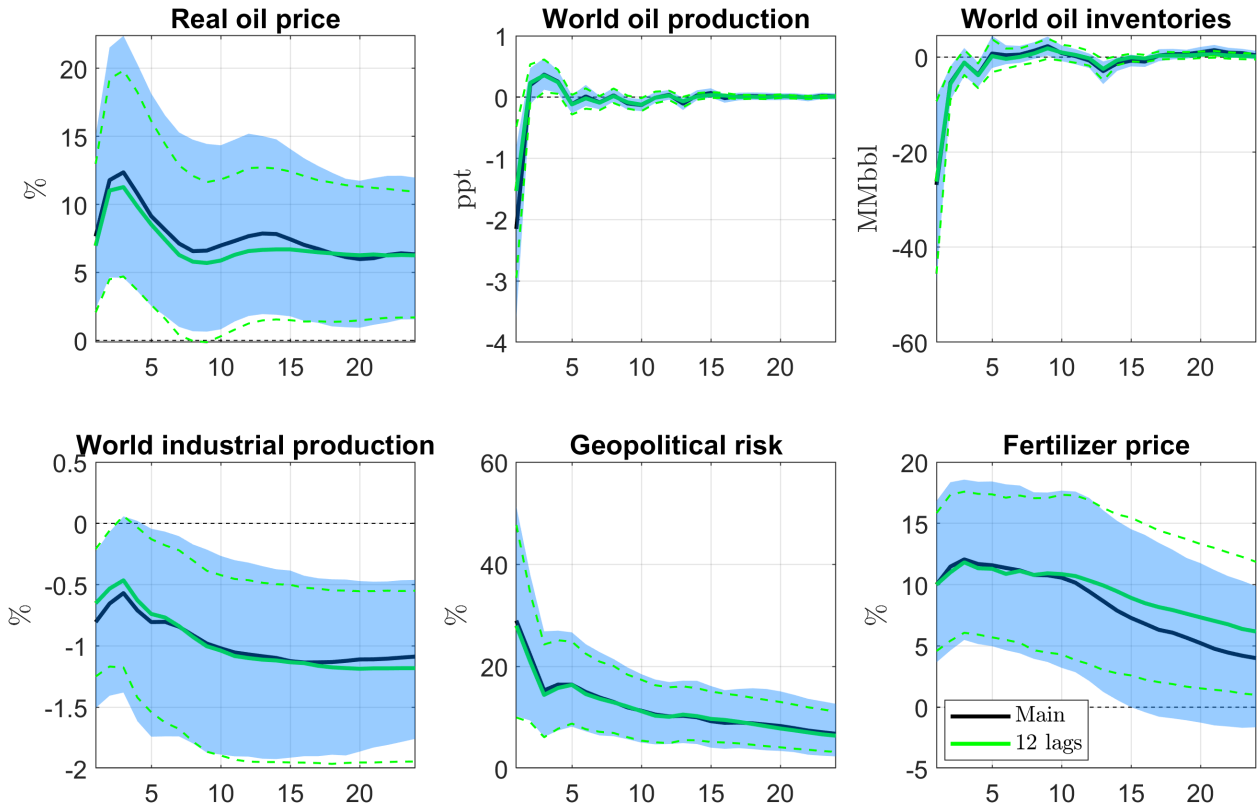


Figure 11: Results from a BVAR(12)

B.7 Robustness: $\mathcal{NW}$ prior

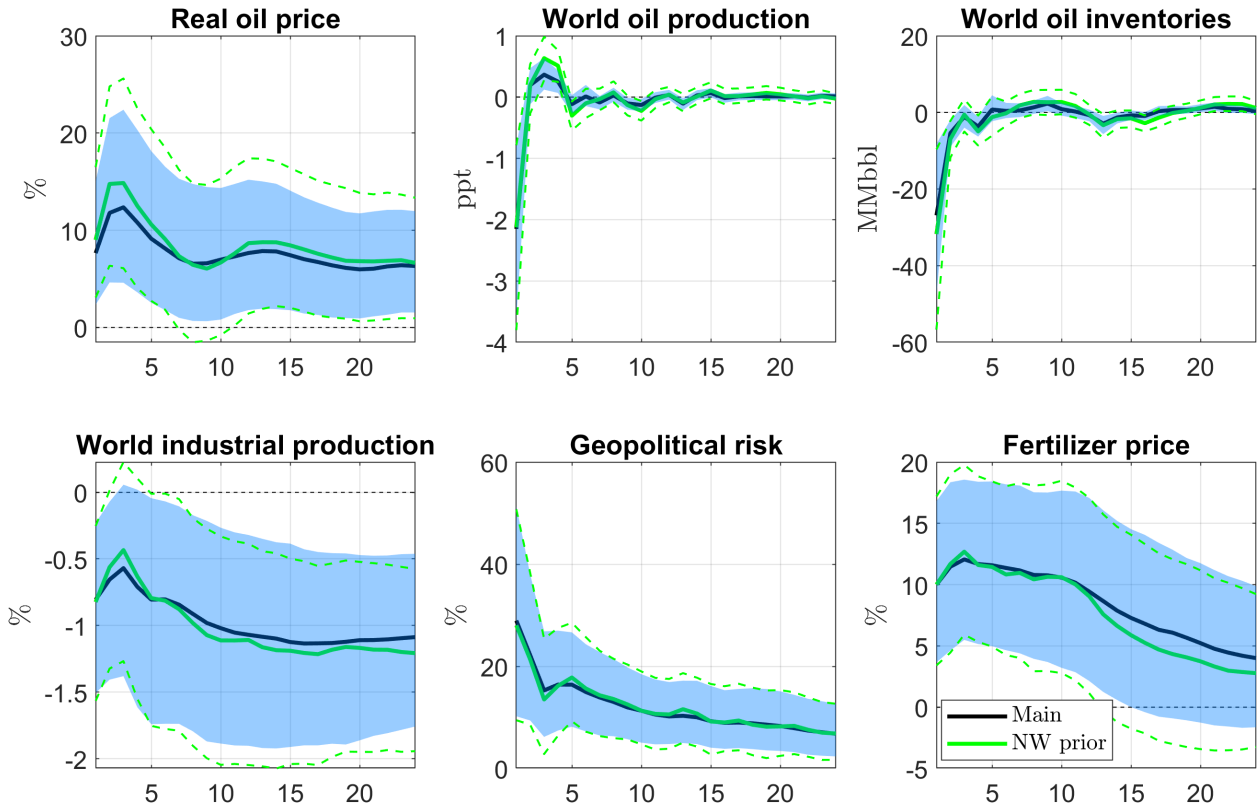


Figure 12: BVAR with Normal-Whishart prior

B.8 Robustness: Normal-Diffuse prior

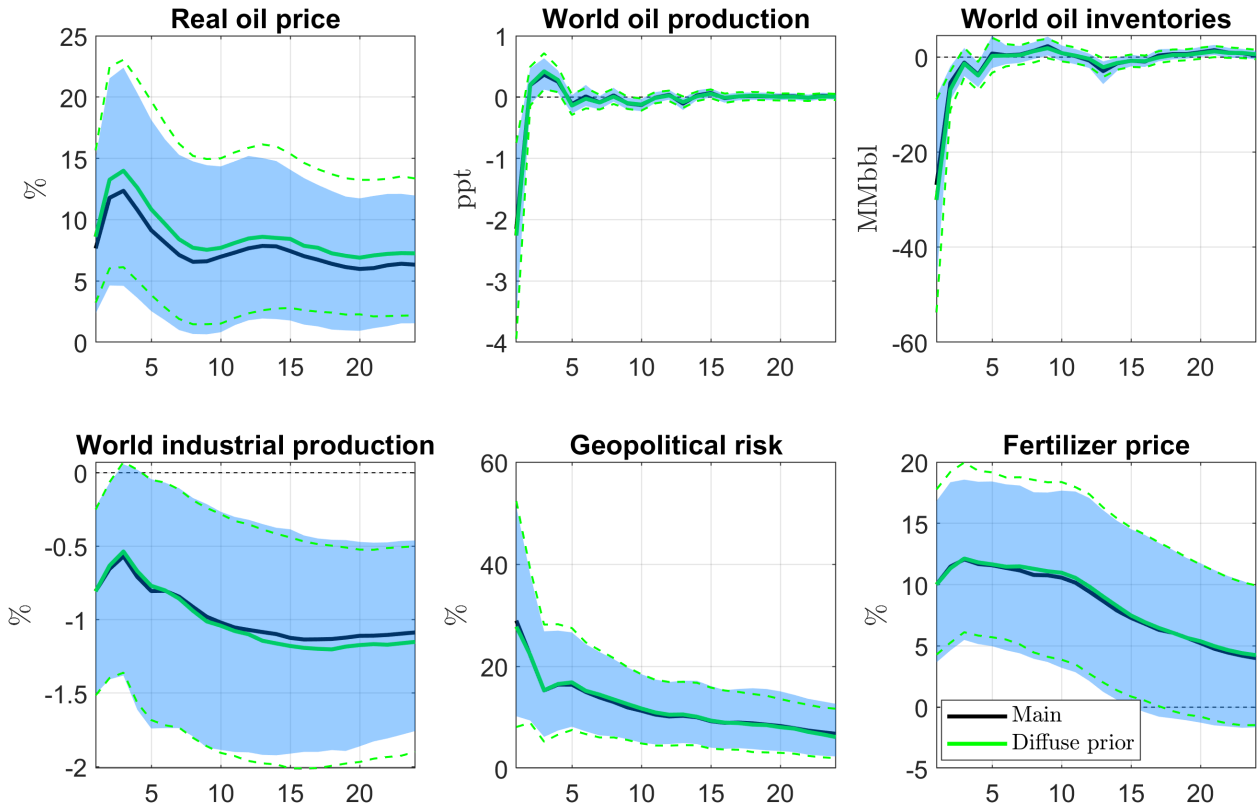


Figure 13: BVAR with Normal-Diffuse prior

B.9 Robustness: $\mathcal{NW}$ diffuse prior

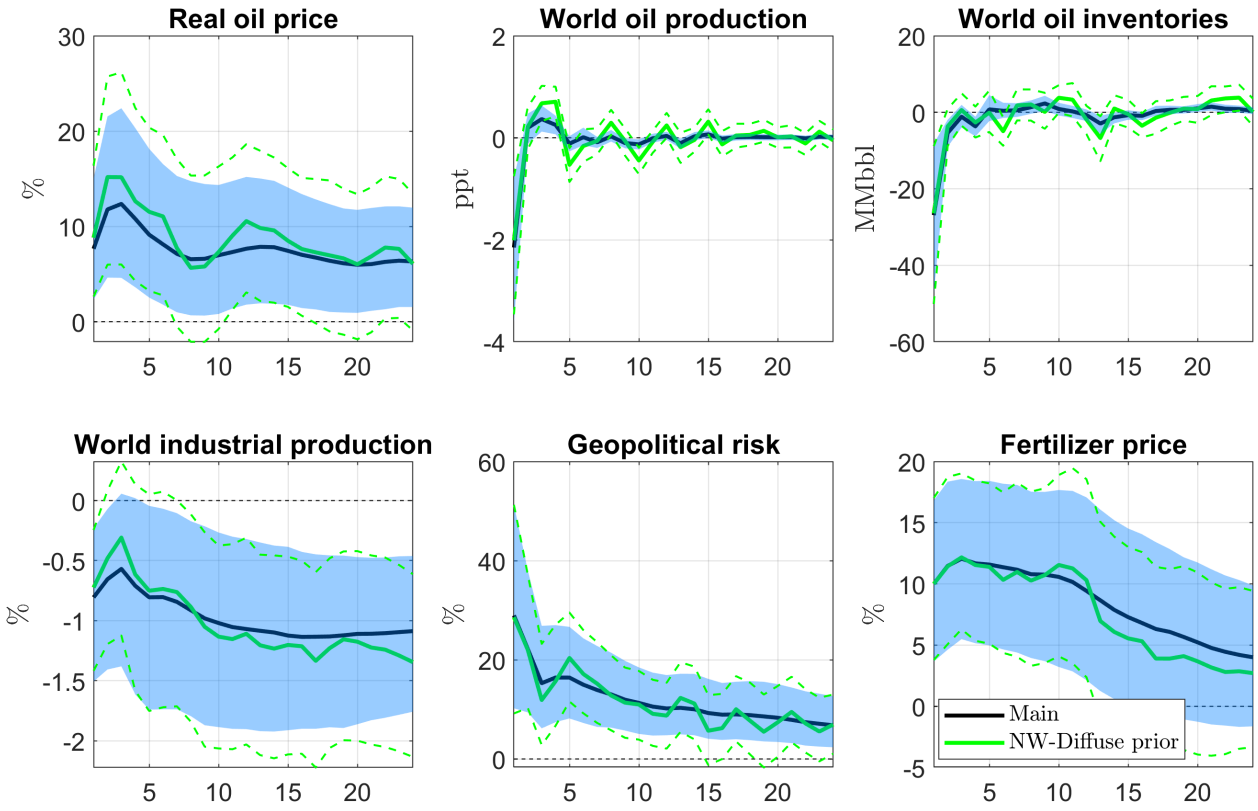


Figure 14: BVAR with Normal-Whishart diffuse prior with $\lambda_1 = 1000$

B.10 Robustness: VAR-OLS

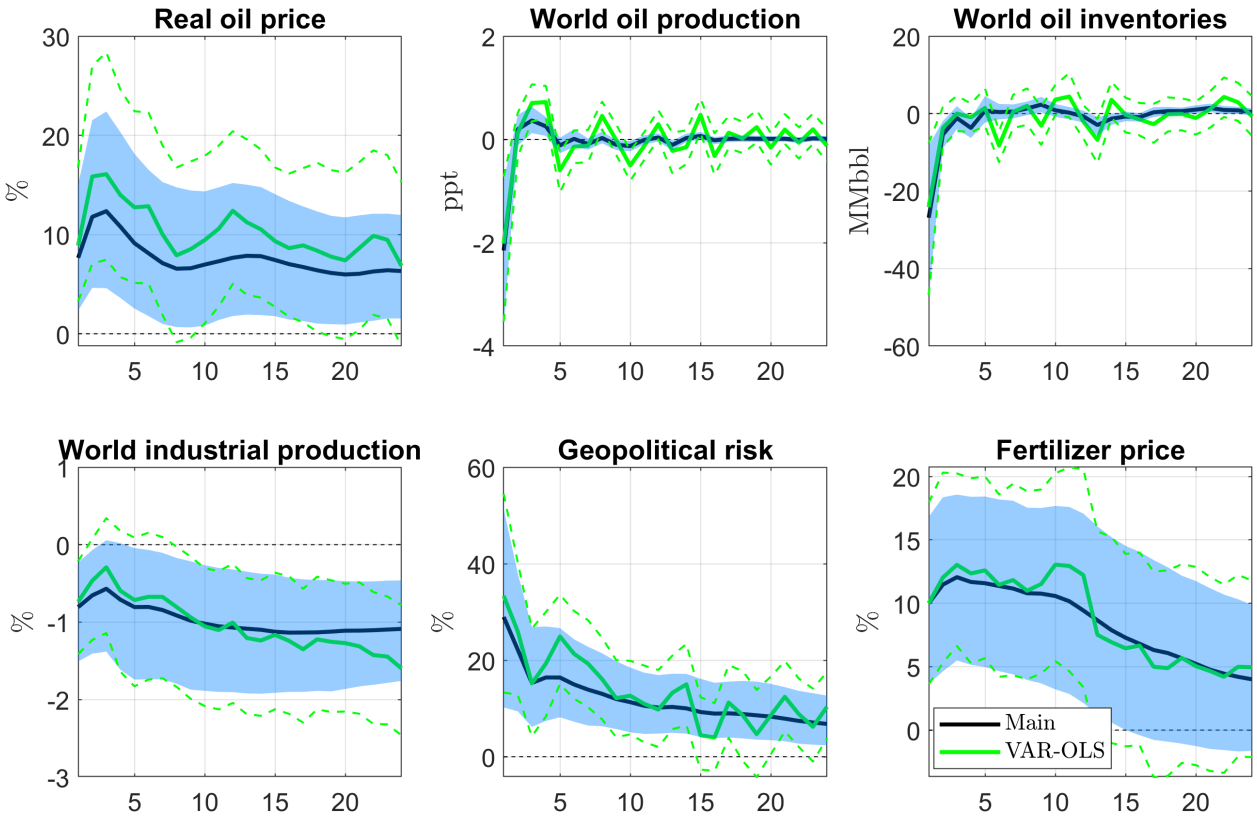


Figure 15: BVAR with $\lambda_1 = 1$ and $\lambda_2 = 2$

B.11 Robustness: Hyperameters optimized

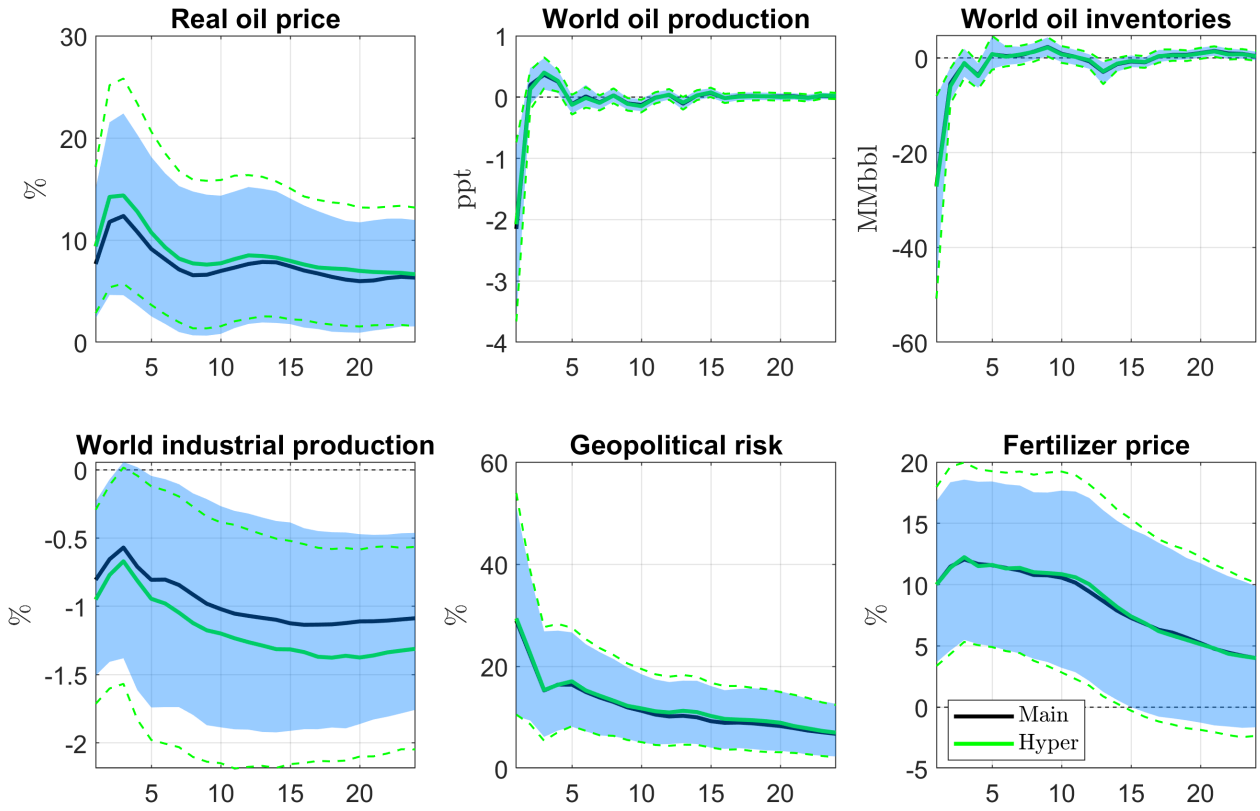


Figure 16: BVAR with hyperparameters optimized by grid search

B.12 Robustness: S_0 identity matrix

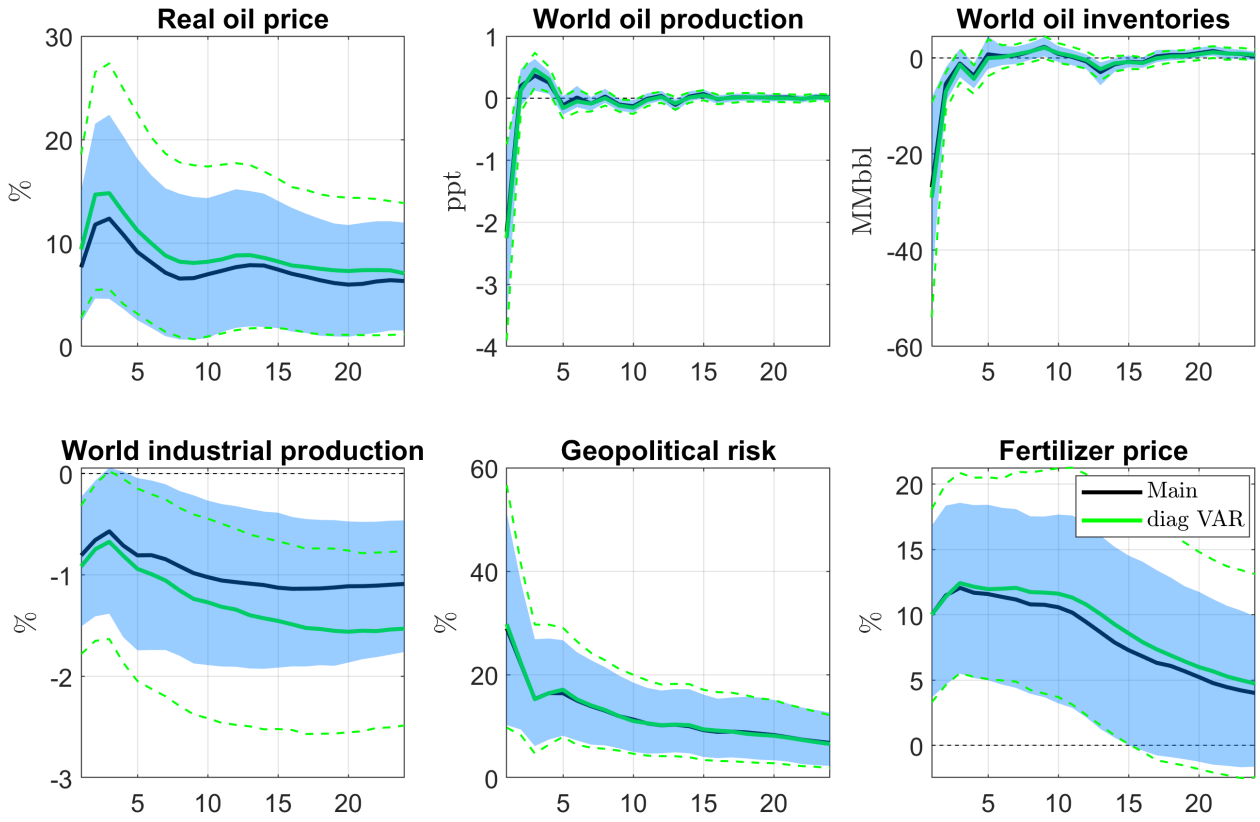


Figure 17: BVAR with S_0 identity matrix

B.13 Robustness: sum-of-coefficients

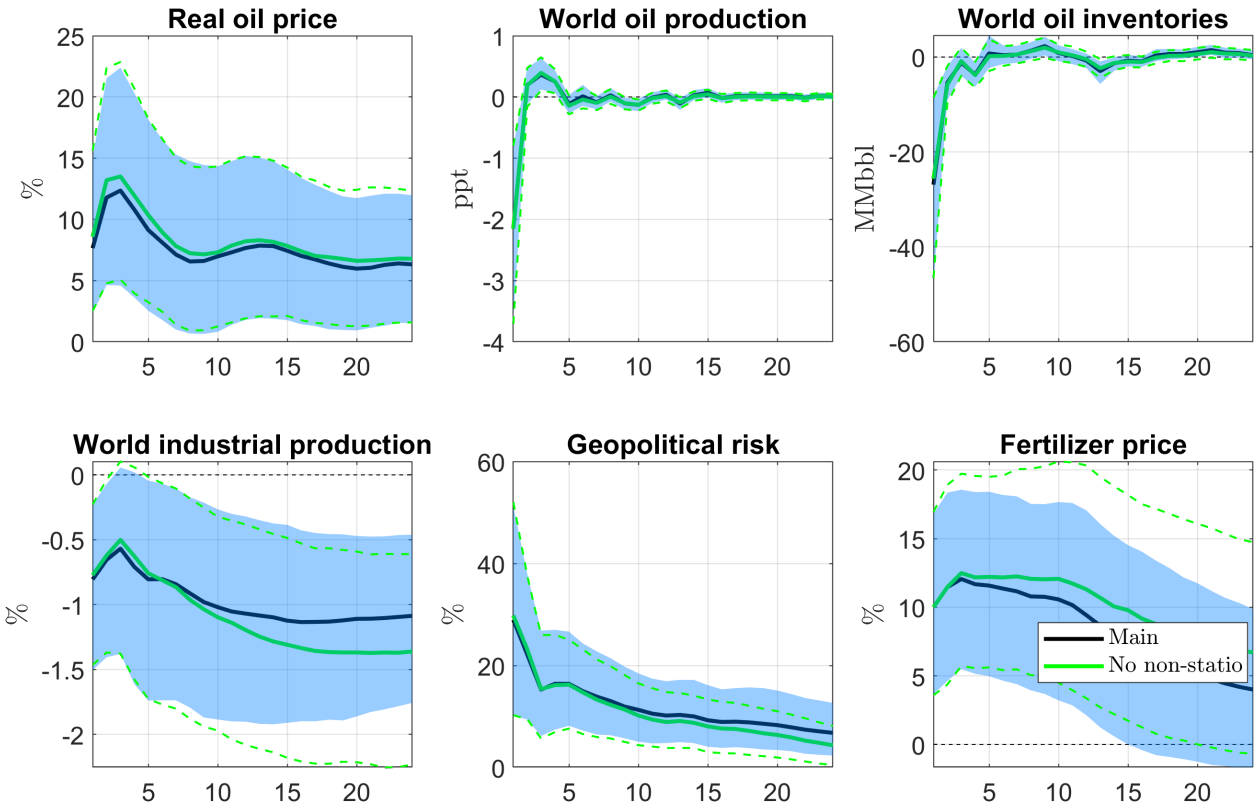


Figure 18: BVAR without sum-of-coefficients

B.14 Robustness: cointegration

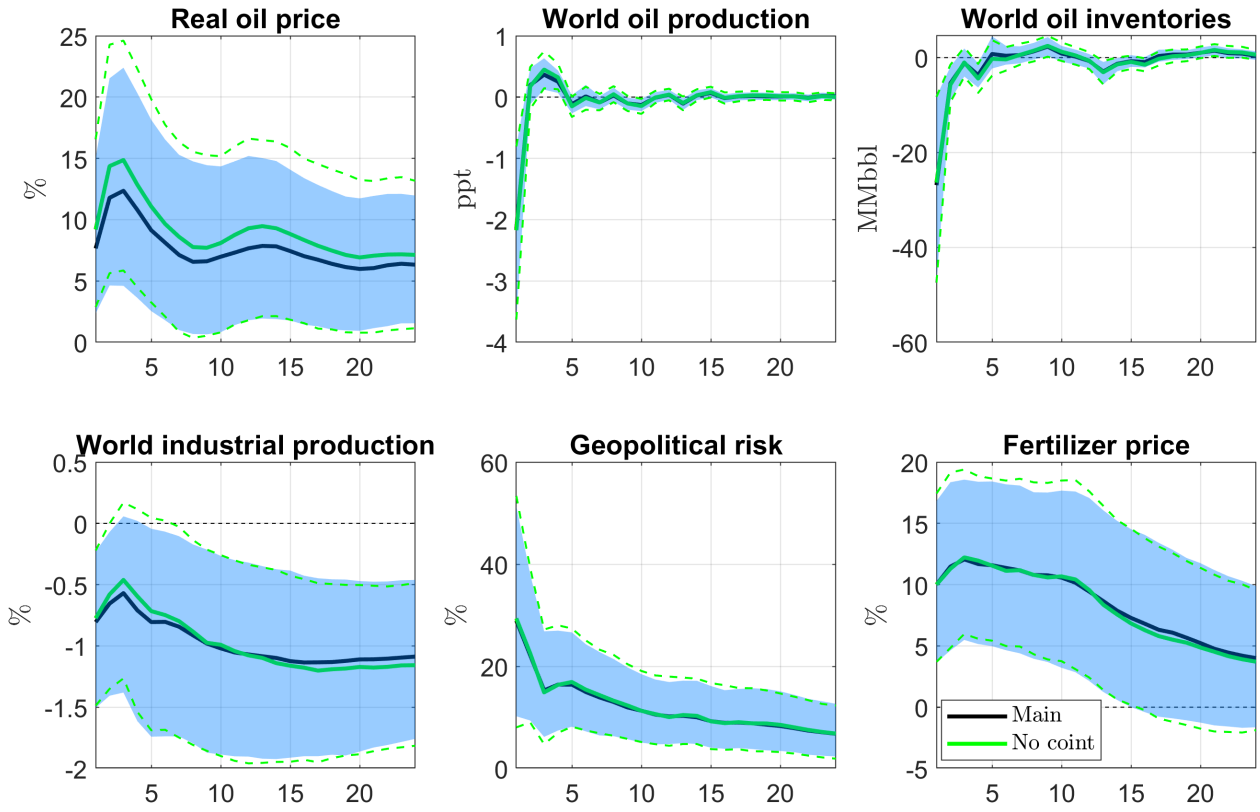


Figure 19: BVAR without dummy prior

B.15 Robustness: linear trend

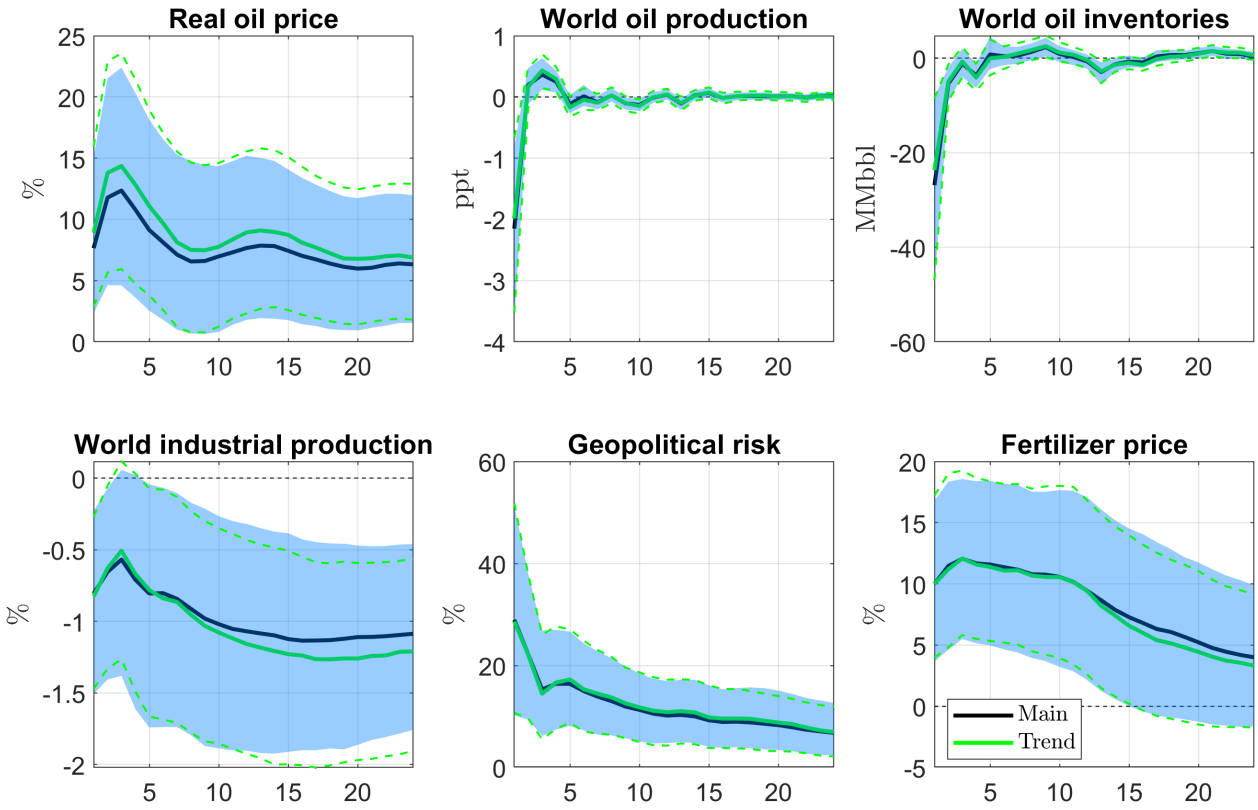


Figure 20: BVAR with a linear trend

B.16 Robustness: no constant

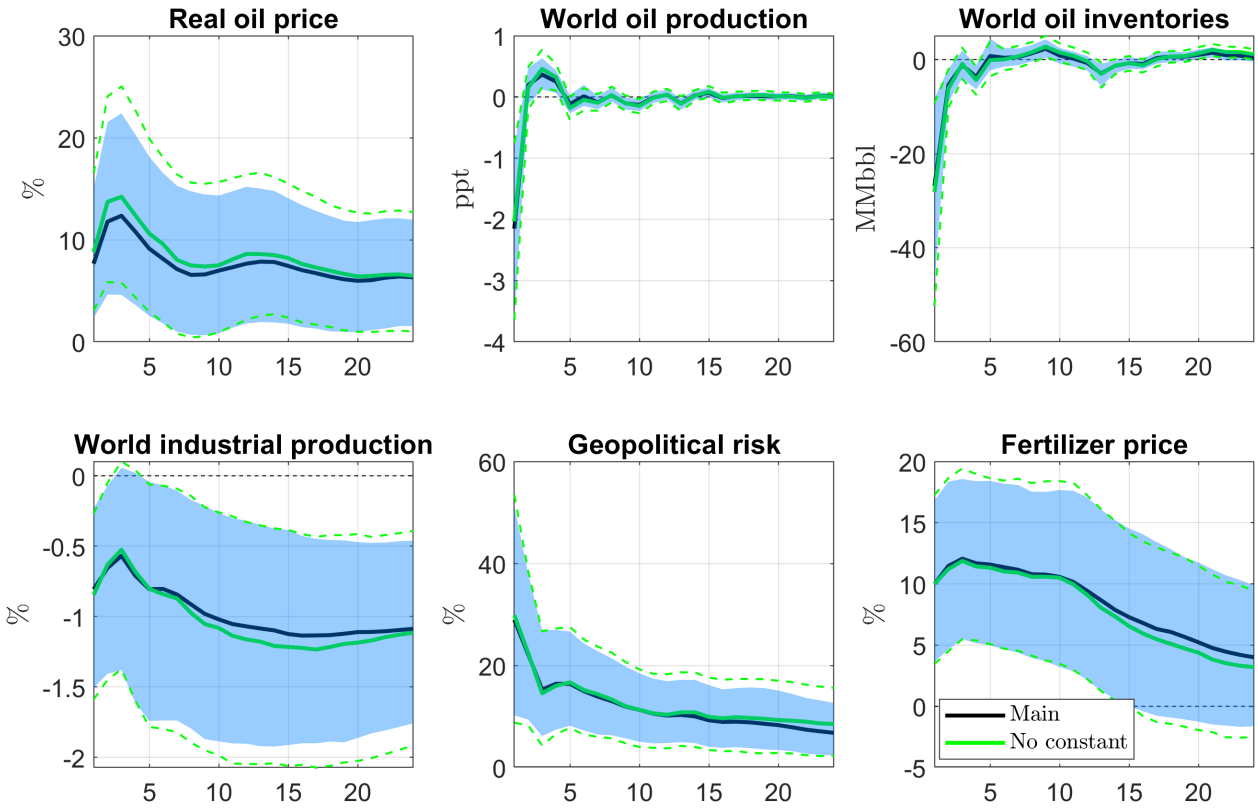


Figure 21: BVAR without a constant

B.17 Robustness: global real economic activity

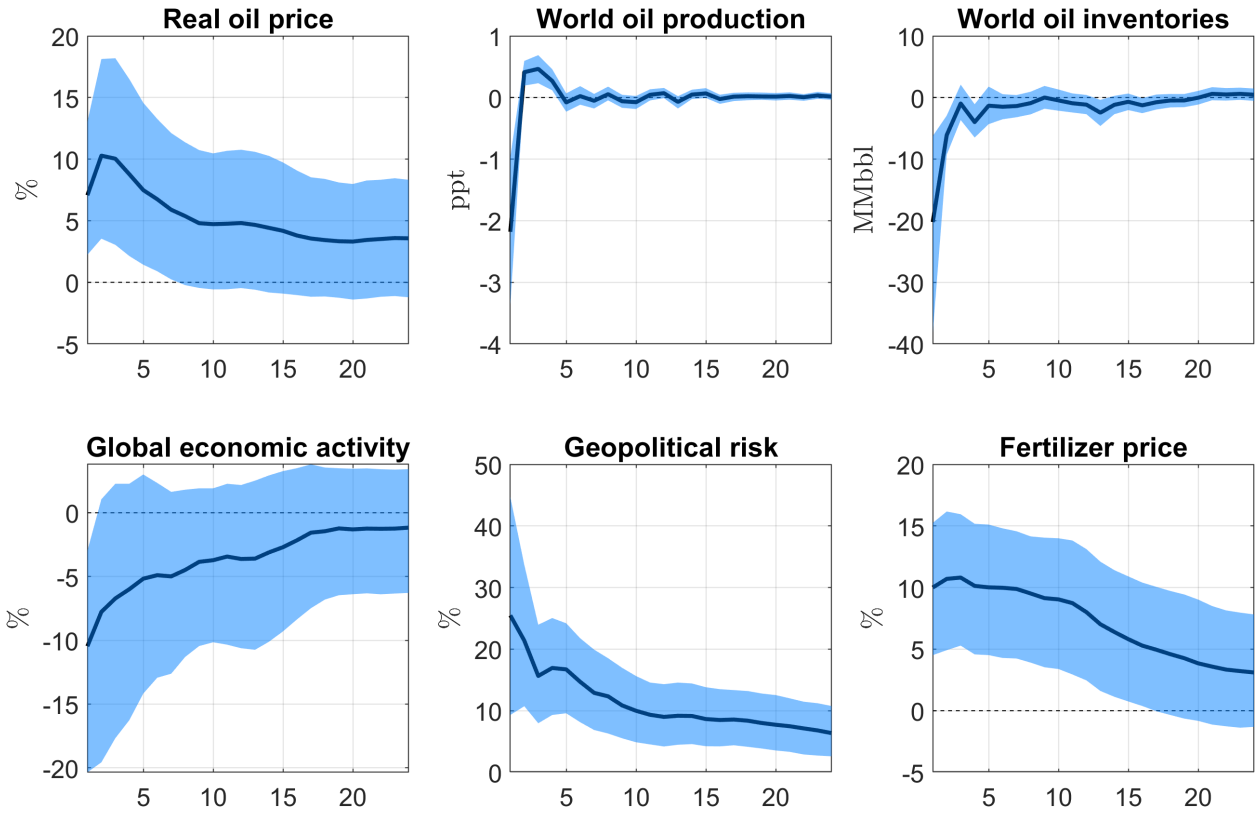


Figure 22: BVAR using index of global real economic activity by by Kilian (2009)

B.18 Robustness: global economic conditions index

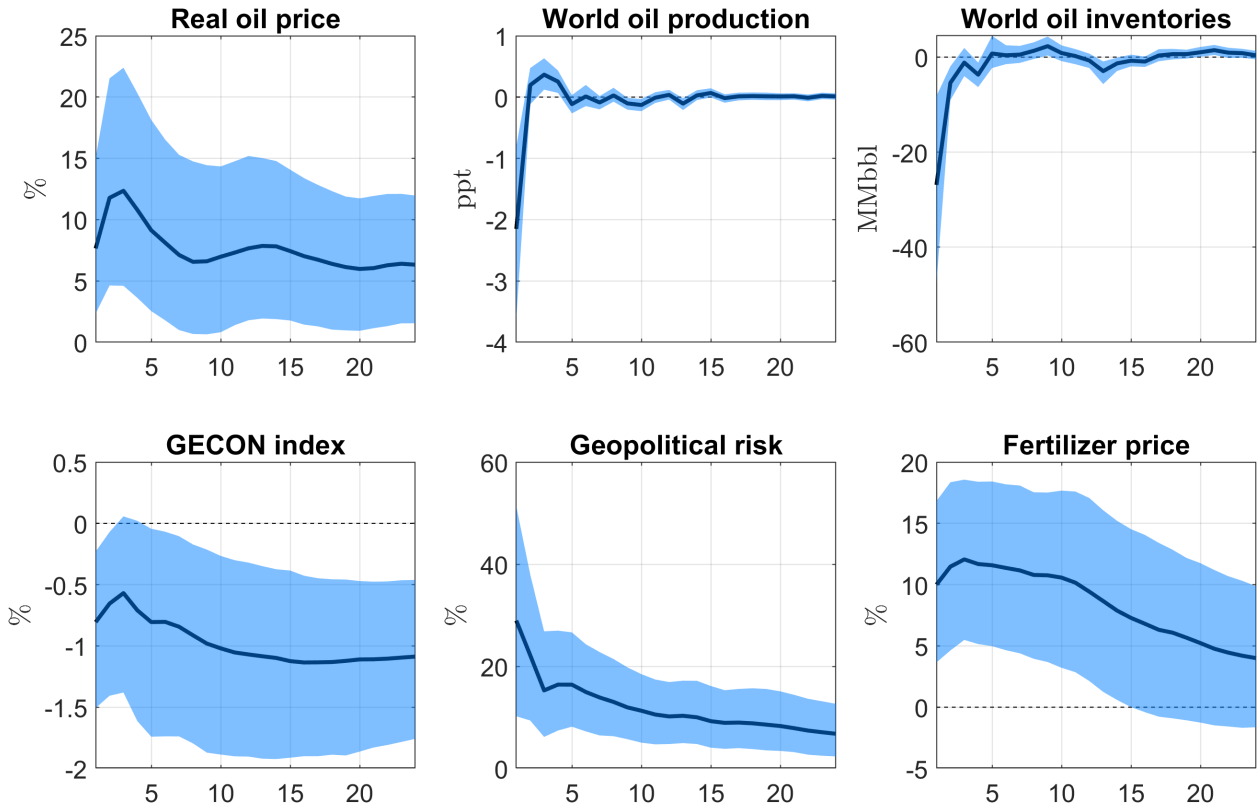


Figure 23: BVAR using global economic conditions index by Baumeister et al. (2022)

B.19 Robustness: natural gas

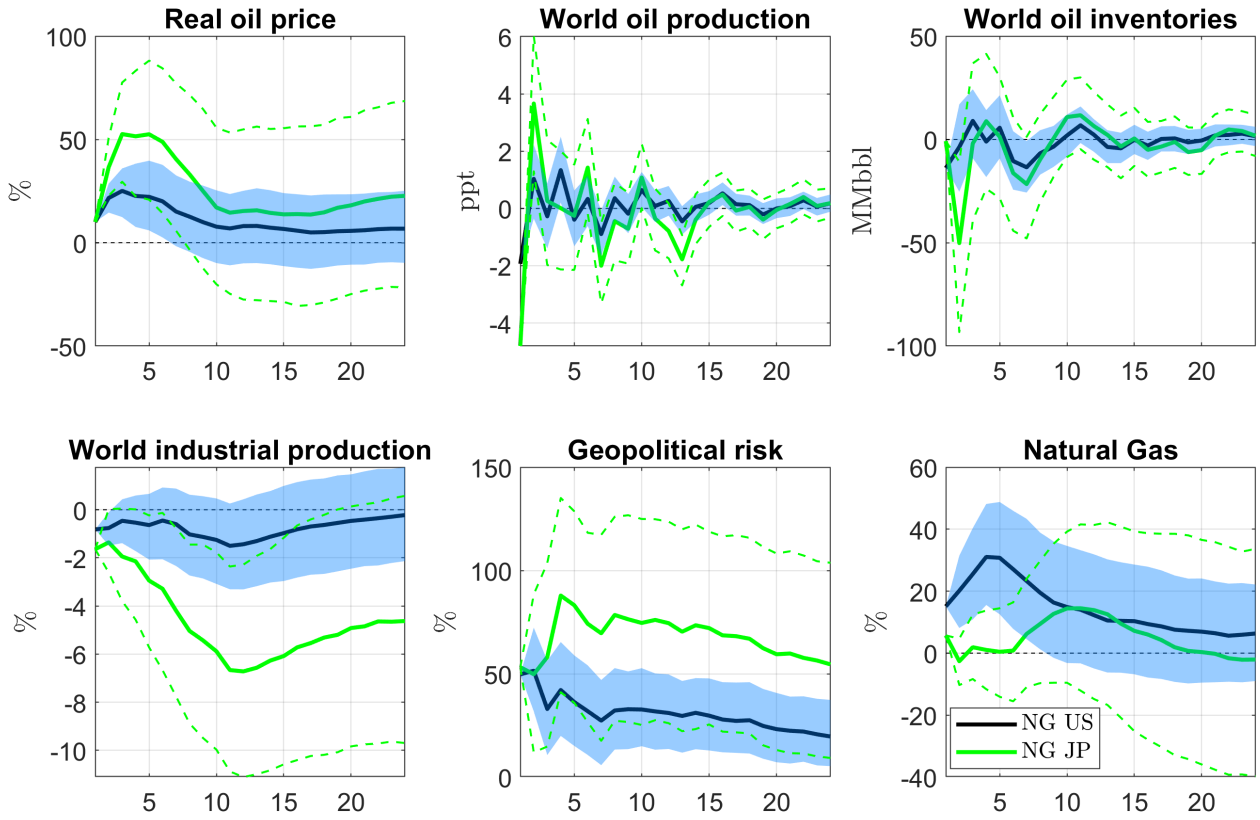


Figure 24: BVAR using natural gas

B.20 Robustness: NEER emerging markets

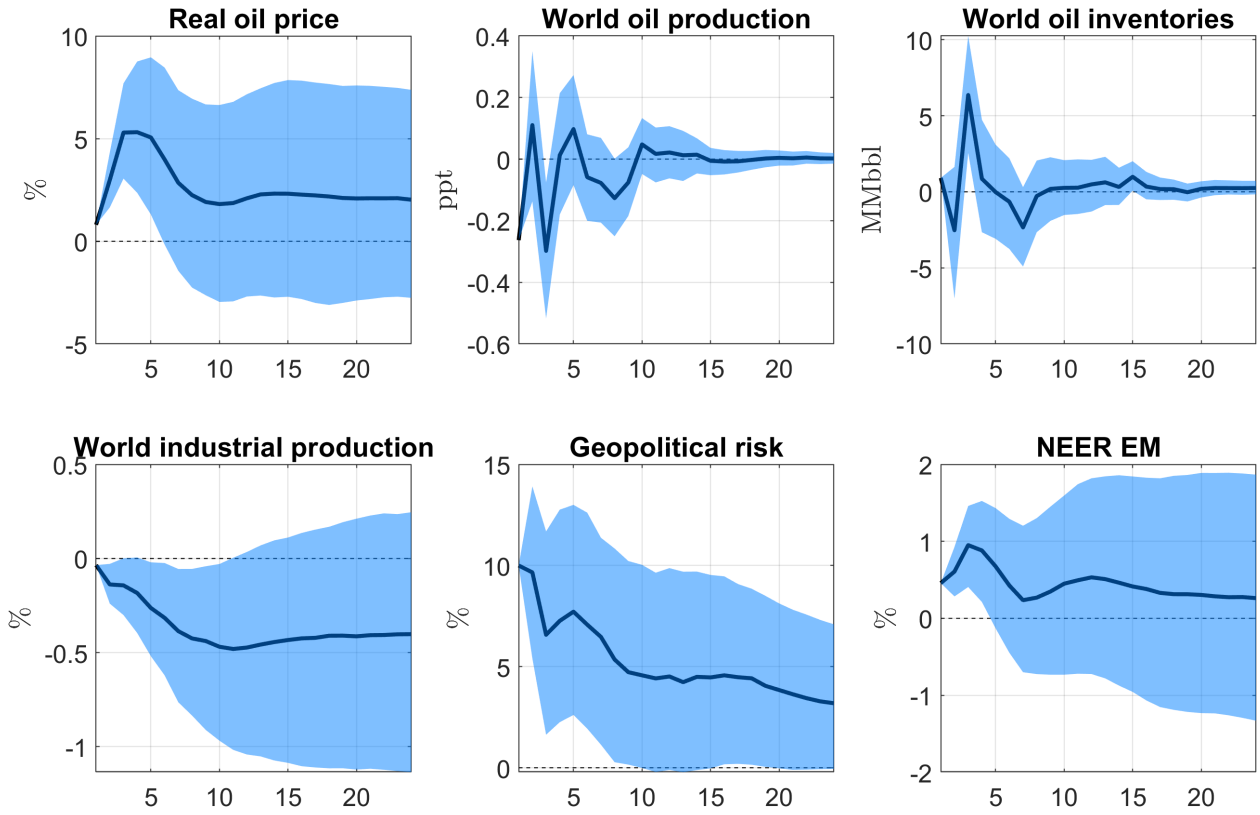


Figure 25: BVAR using NEER emerging markets

B.21 Robustness: NEER advanced economies

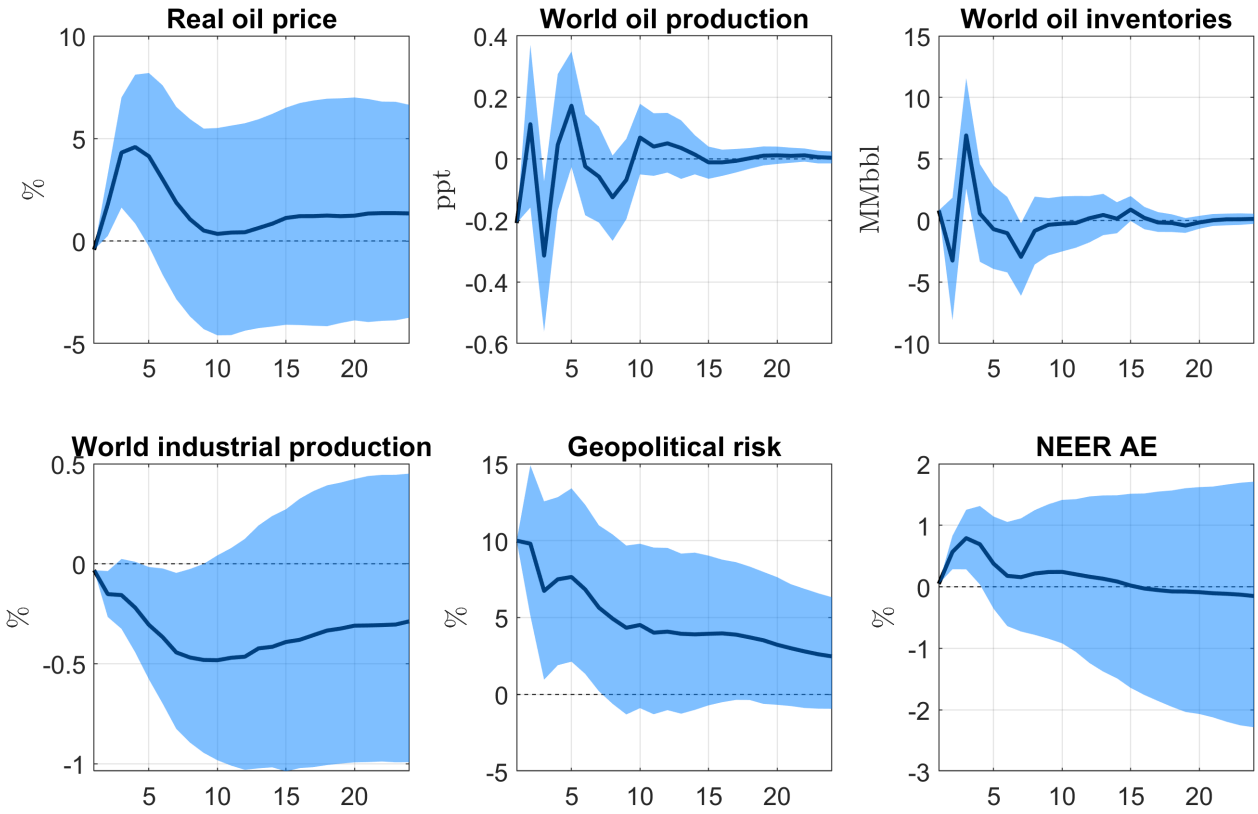


Figure 26: BVAR using NEER advanced economies

B.22 Robustness: NEER United States

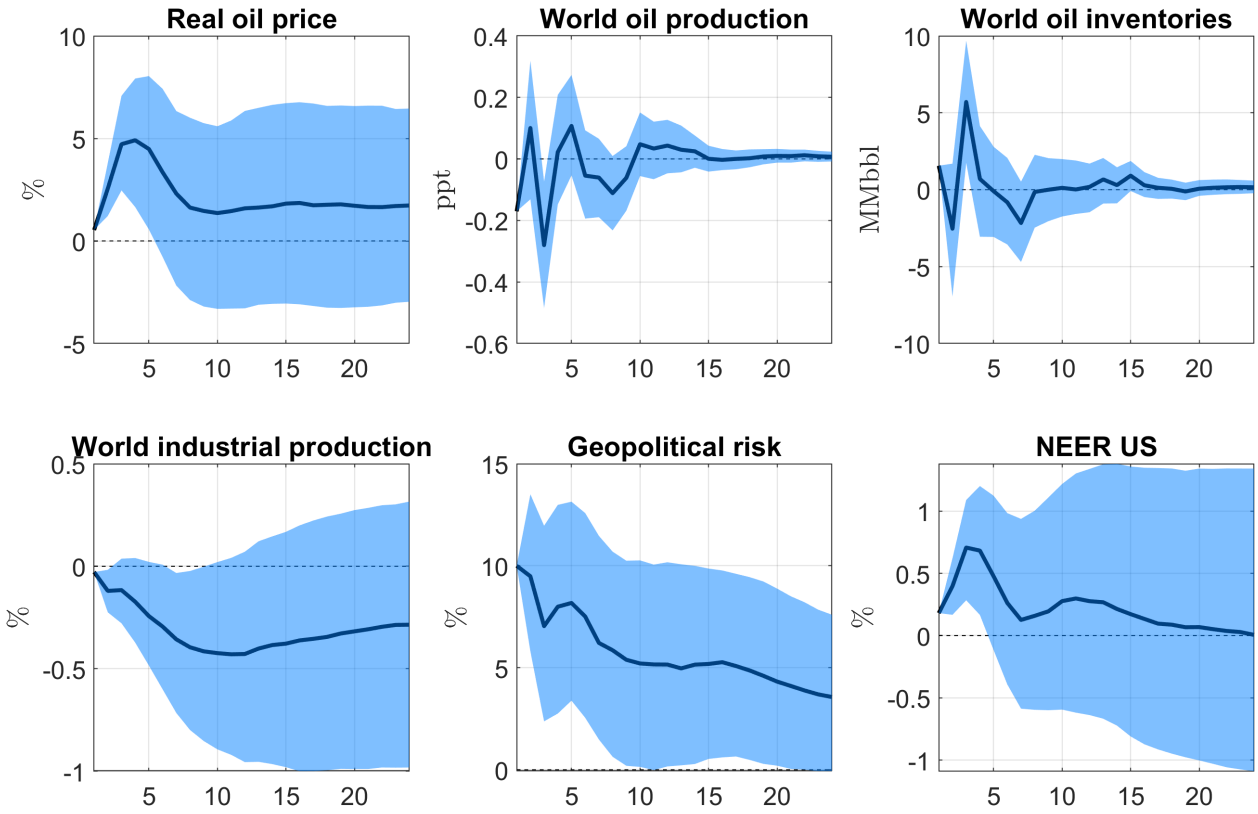


Figure 27: BVAR using NEER US

B.23 Robustness: NEER Euro Area

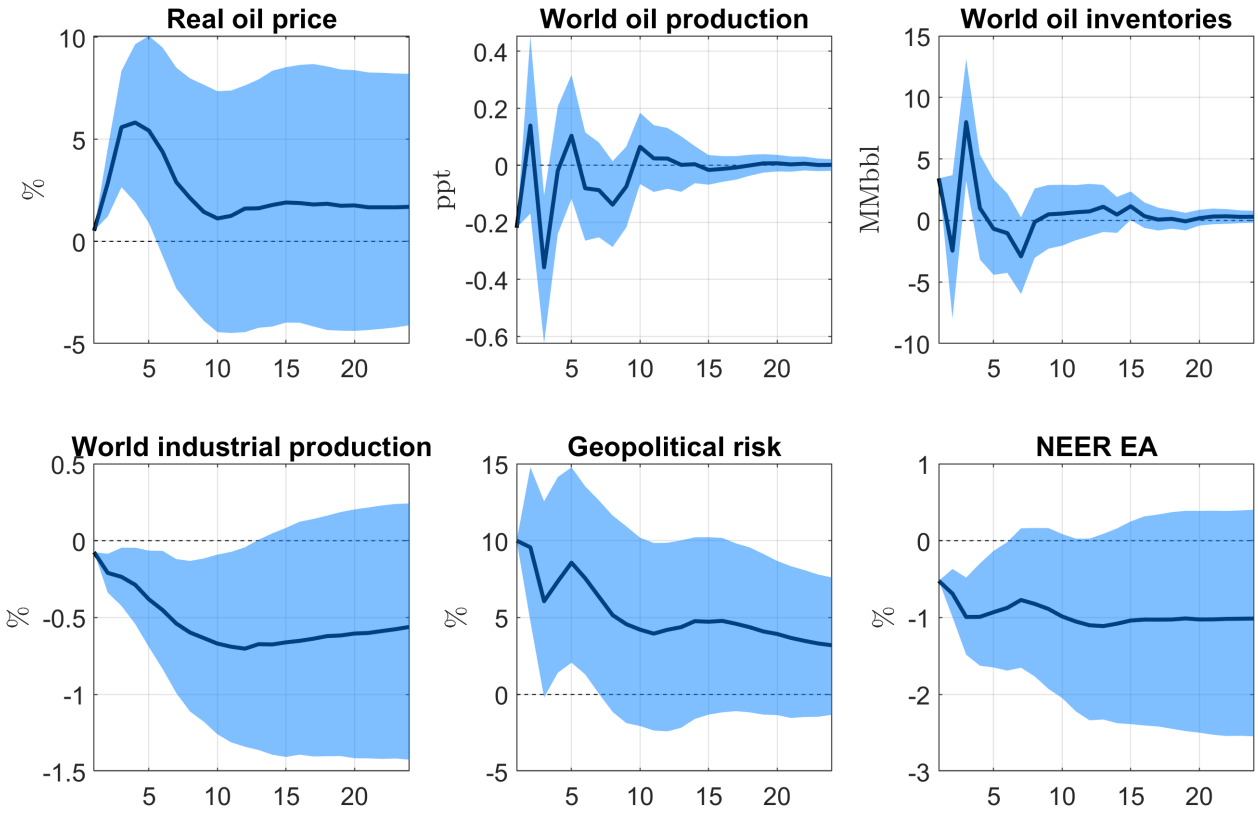


Figure 28: BVAR using NEER EA

B.24 Robustness: NEER United Kingdom

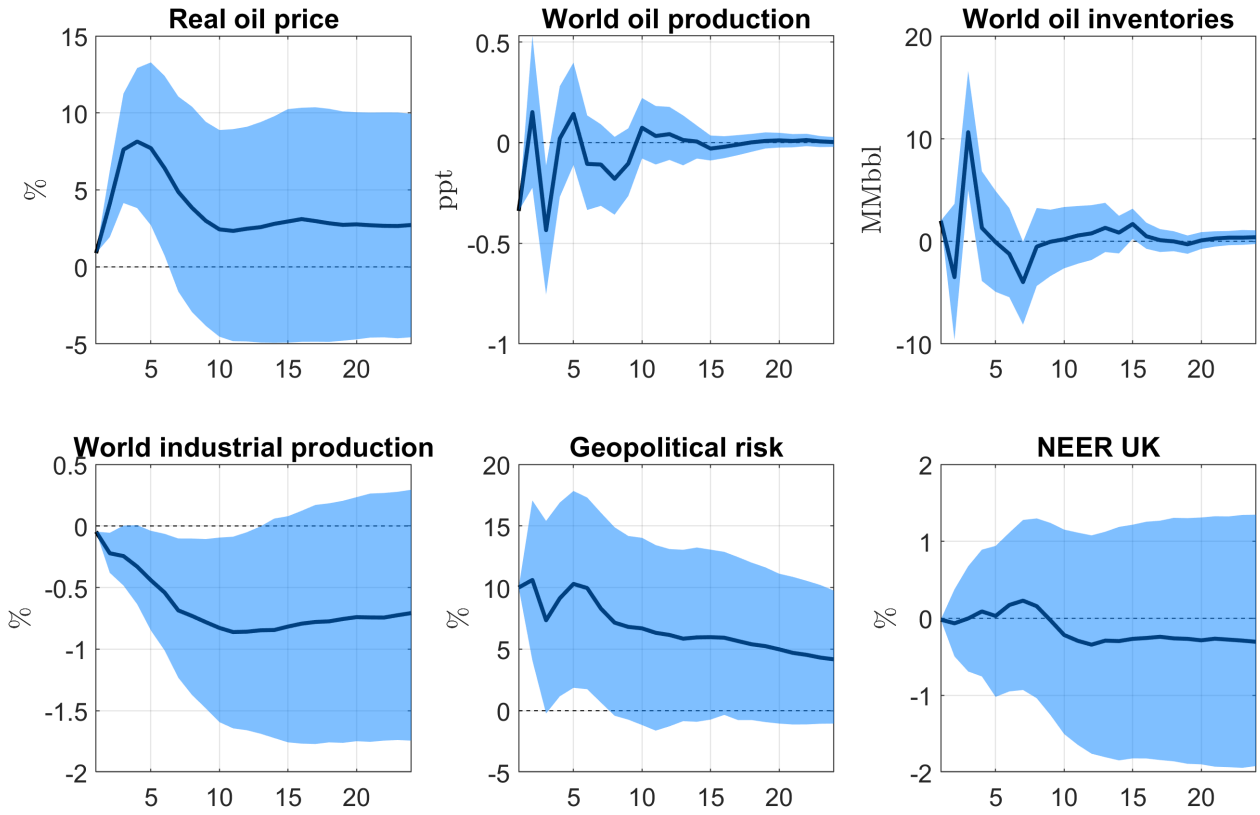


Figure 29: BVAR using NEER UK

B.25 Robustness: NEER Japan

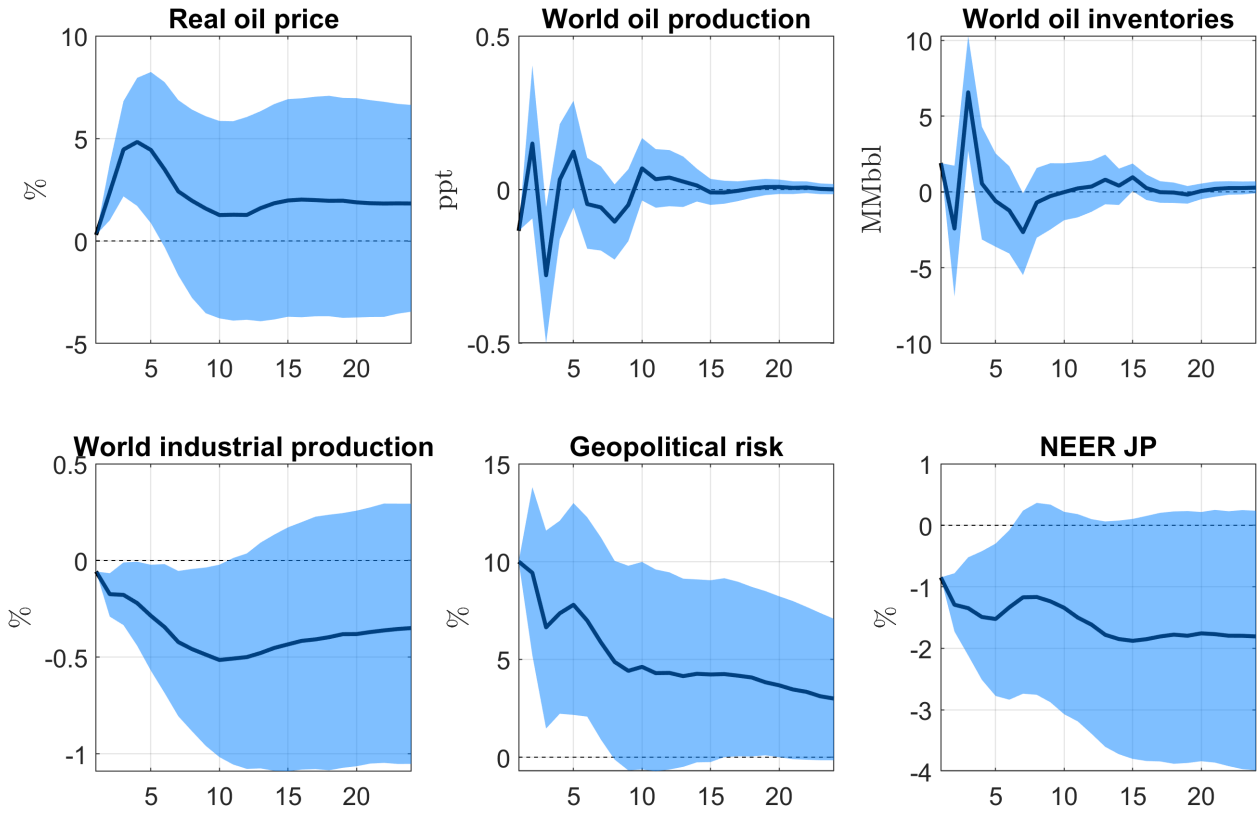


Figure 30: BVAR using NEER Japan

B.26 Robustness: NEER Switzerland

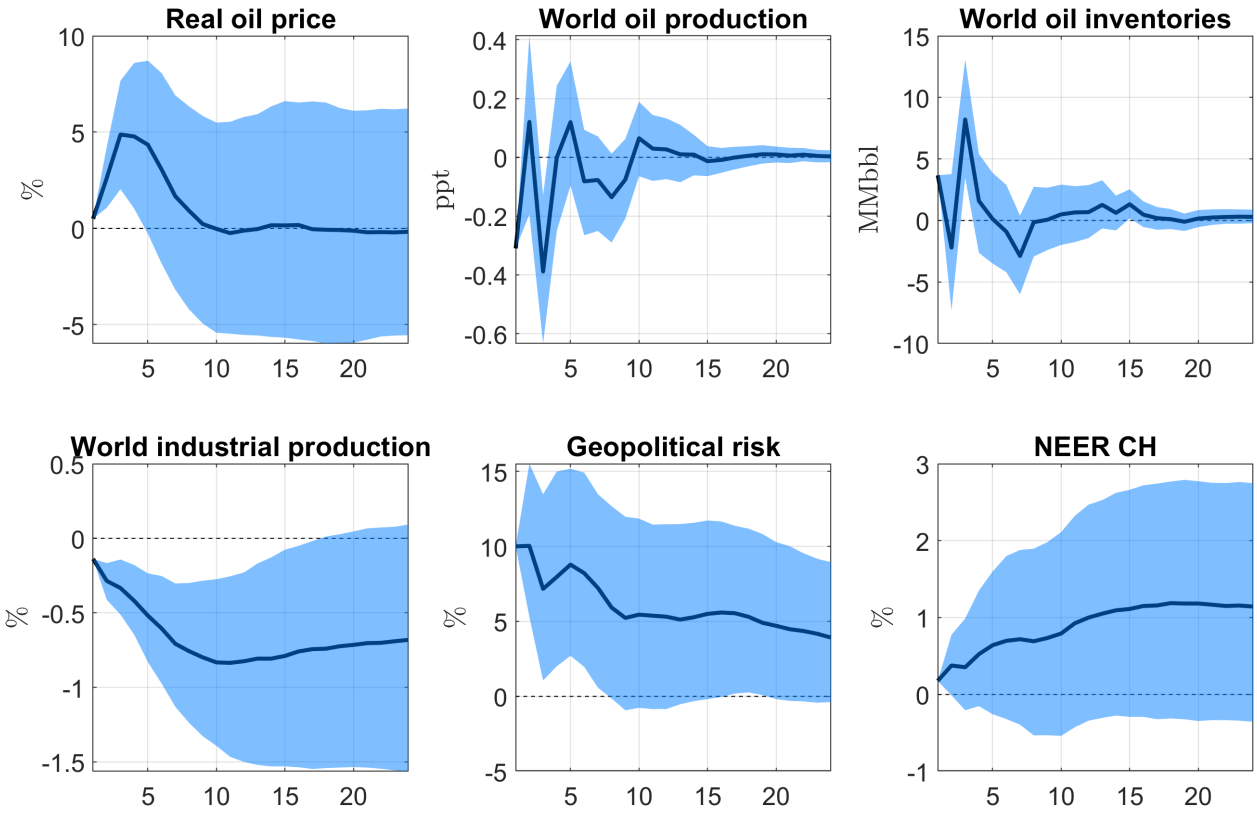


Figure 31: BVAR using NEER Switzerland